Such false detections can be minimized by adjusting the validation parameters. For example, it is possible to adjust the resolution used for sector division or to reduce the minimum number of positive sectors in order to detect half cylinders. However, such adjustments lead to an increase in false detections, which is described in more detail in the evaluation of half cylinder detection.

Since there are no false detections and only one misdetection in 1000 · 16 cylinder detections, the circle detection parameters have been chosen well enough for an evaluation of the circle detection.

Detected cylinder axes

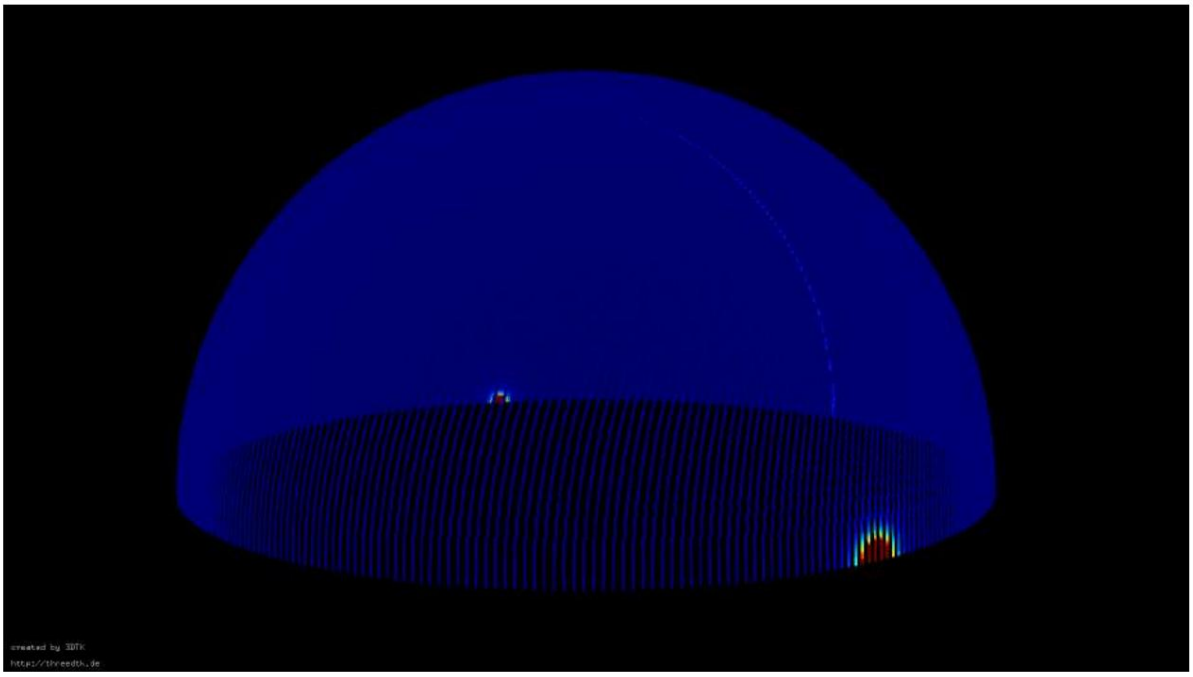


Figure 5.4: Representation of the accumulator for detecting the cylinder axes, where blue represents unincremented cells and red represents cells with maximum accumulation.

Before the actual evaluation of the circle detection begins, it is useful to briefly analyze the detected cylinder axes and their errors to understand which systematic errors are introduced into the circle detection process. This is important because even small errors in the cylinder axes can affect the projection and thus complicate circle detection.

As before, a left-handed point cloud of the model shown in Appendix 6.1 is used as input. The perfect axis of each cylinder is therefore [0, 1, 0] or [0, -1, 0].
Because of this axis, two maxima are calculated at the edge of the sphere in Hough space.
The Hough space is again shown graphically in Figure 5.4, where blue dots correspond to accumulator cells with low values and red dots to maxima. Since the cylinder detection does not know how many cylinder axes occur in the point cloud,

Despite the model having a single, defined axis, multiple axes can be detected. New circle detections are only stopped when enough points have been assigned to cylinders, the maxima in Hough space have been processed, or the user-defined maximum number of axes (in this case, 10) has been exceeded.

Detected axis	Number of detections	Error angle difference [°]
[0.003148 -0.999992 0.002356]	15	0.229183
[0.003148 0.999992 0.002356]	80	0.229183
[0.003148 -0.999995 0.000785]	3328	0.181185
[0.003148 0.999995 0.000785]	12576	0.181185

Table 5.3: Detected cylinder axes and their error angles

Table 5.3 clearly shows that four cylinder axes were found for the model in 1000 runs, with each pair of cylinder axes originating from the two maxima of Hough space. The two cylinder axes with an error angle difference of 0.181° are detected for approximately 99.4% of the cylinders, while the two axes with the larger angle difference of 0.229° are detected for only 0.6% of the runs. However, all four axes exhibit only a small deviation from reality and could be further improved by using a larger accumulator, more frequent random calculations of the great circles, or adding more sample points.

Radius determination errors

At the beginning of the circle detection evaluation, the detected radii are again considered first, followed by validation of the circle center determination. An important aspect of the evaluation in this case is how errors in the cylinder axis affect and propagate to the circle detection.

Analogous to the evaluation of circle detection with a perfect cylinder axis, all detected radii are shown in a histogram (5.5). Since the errors of the calculated axes are small, no difference can be visually discerned in the histogram between the representation of the radii with a perfect axis (see Figure 5.1) and the detected axis. It is also barely noticeable that the first bar for the circles with a radius of 2.5 mm is slightly shorter, because instead of the expected 4000 cylinders with a radius of 2.5 mm, only 3999 were detected. As mentioned at the beginning, this is due to the small number of lateral surface points. Because of the error-prone axis and thus the error in the projected points, as well as the possibility of poor circle detection by the RHT, it is significantly more difficult to correctly detect cylinders with few lateral surface points. However, no further difference is discernible in the histogram.

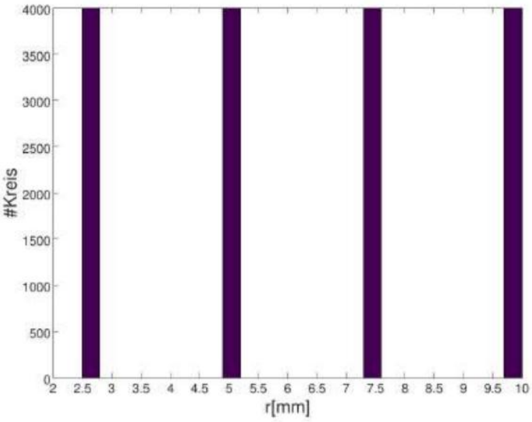


Figure 5.5: Representation of the found radii

For this reason, the errors of the different radii will be considered individually next. The errors in histograms are shown in Figure 5.6. They already show... A difference was detected between circle detection with a perfect and with a faulty axis. will be. Instead of the four error bars very close together in Figure 5.2 with For each value of 1000, a different number of errors are now displayed. This is due to the fact that... together, that now a distinction is made not only between the radius and height of the cylinder, but also how large the actual error angle is between the absolute and detected values The axis is the larger the error of the axis, the greater the inaccuracy of the detection.

Radius [mm]	mean Radius [mm]	Mean absolute Error [mm]	Standard deviation [mm]
2.5	2.4997	7.3633e-04	7.6322e-04
5.0	4.9995	5.7031e-04	4,7695e-04
7.5	7.4990	9.5295e-04	1.9202e-04
10.0	9.9984	1.5972e-03	1,5551e-04

Table 5.4: Mean, absolute error and standard deviation of the four radii

In contrast, the error of the mean value has become smaller compared to detection with an optimal cylinder axis (see Table 5.4). However, this is only due to the fact that, as a result of the Both positive and negative radius errors can occur on an erroneous axis. From this The reason used in the following evaluations will always be the mean absolute error of the measurement. This shows clearly the average error of the radius. is significantly larger than with a given perfect cylinder axis. Nevertheless, the detected Radius very close to the actual absolute value.

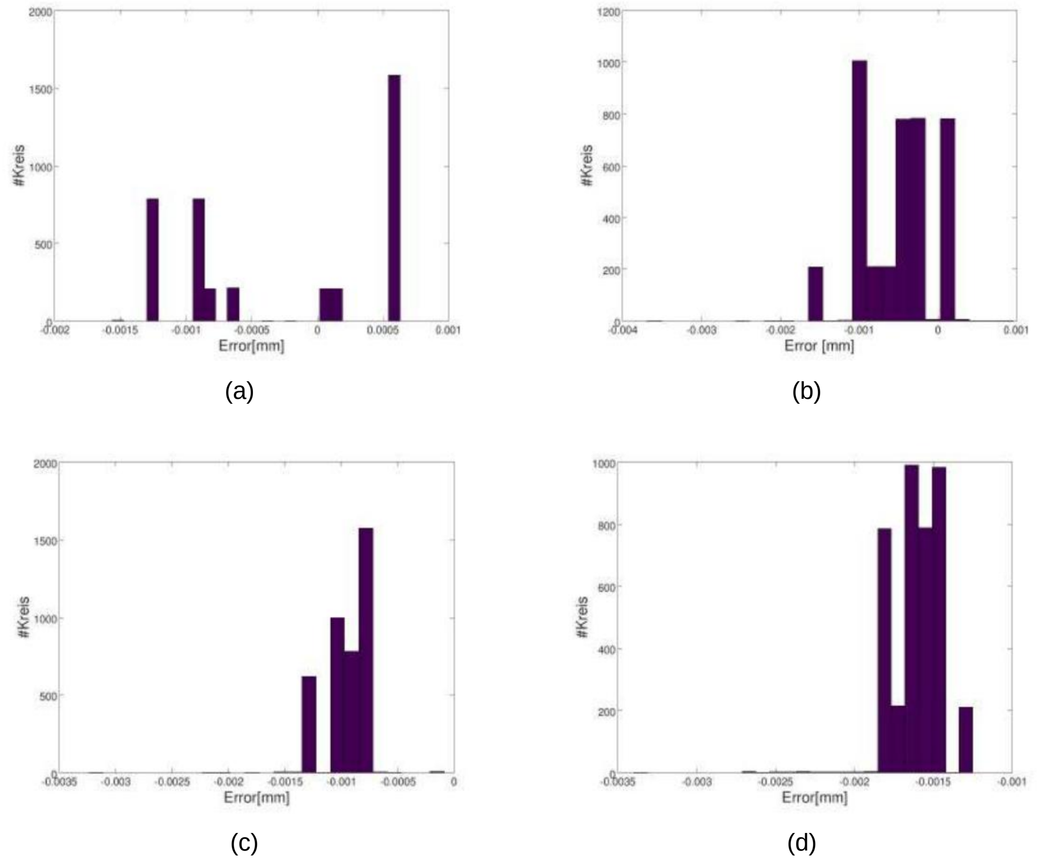


Figure 5.6: Errors of individual radii: (a) Errors for circles with radius 2.5; (b) Errors for circles with radius 5.0; (c) Errors for circles with radius 7.5; (d) Errors for circles with radius 10.

Another trend can be observed based on the standard deviation. The standard deviation, and therefore the spread of the radius, is greater the smaller the radius.

This supports the claim that the detection of circles with few erroneous projected lateral points is significantly more difficult and therefore also shows larger variations.

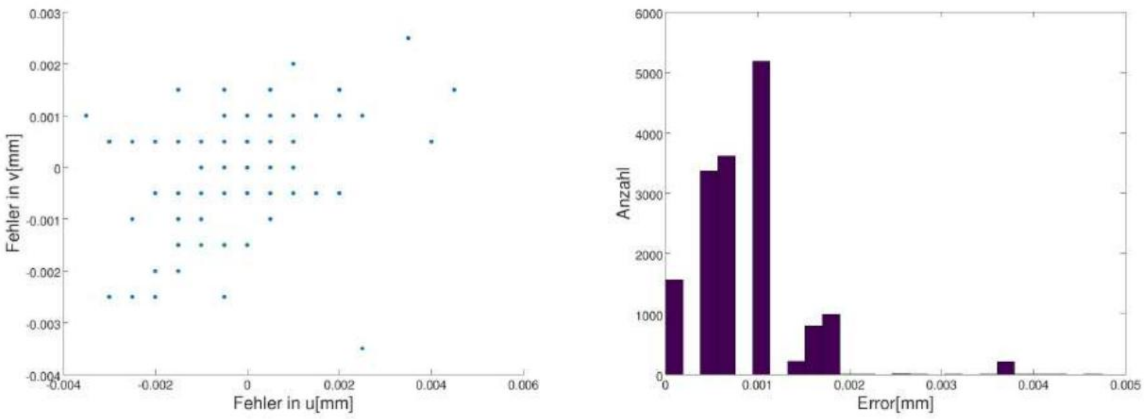
If there are very few points available, improvement using the least-squares approach is also significantly more difficult, as errors are less able to cancel each other out due to the small number of measurement points.

Overall, however, it can be stated that the radii are successfully detected despite the faulty axis.

Position determination error

After evaluating the radius measurement, closer attention is now paid to the detected position of the cylinder in three-dimensional space. Analogous to the evaluation of the position of the detected cylinders for a perfect axis, the three-dimensional point is projected onto the cylinder axis along this axis and then compared with an absolute target value.

A visual representation of all projected points in a plot is omitted because the inaccuracies in determining the circle centers are very small. Instead, the errors of the circle centers are calculated using the absolute coordinates and shown in Figure 5.7. In the figure on the left, all 15,999 errors of the circle centers on the plane are plotted in the projected coordinates [u, v]. Due to the least-squares approach and the simulated point cloud, only about 90 unique error pairs are found for the projected positions. In the figures on the right, the error distances of each calculated position are summarized in a histogram. The maximum distance error is 0.00474 mm, and the average error is 0.000889 mm.



(a) Error of the circle centers in (u, v)

(b) Distance error of the circle centers

Figure 5.7: Errors in determining the center of a circle

Figure 5.7 clearly shows that while the calculated positions do not correspond exactly to the absolute values, the error is very small. Table 5.5 again presents the mean values of the circle centers, the mean distance error, and the standard deviation, using the same cylinder numbering as in the previous trial (shown in Figure 5.3). This demonstrates that there is an error in the projected positions, but it is very small and always within an acceptable range.

Evaluation of radius and position determination

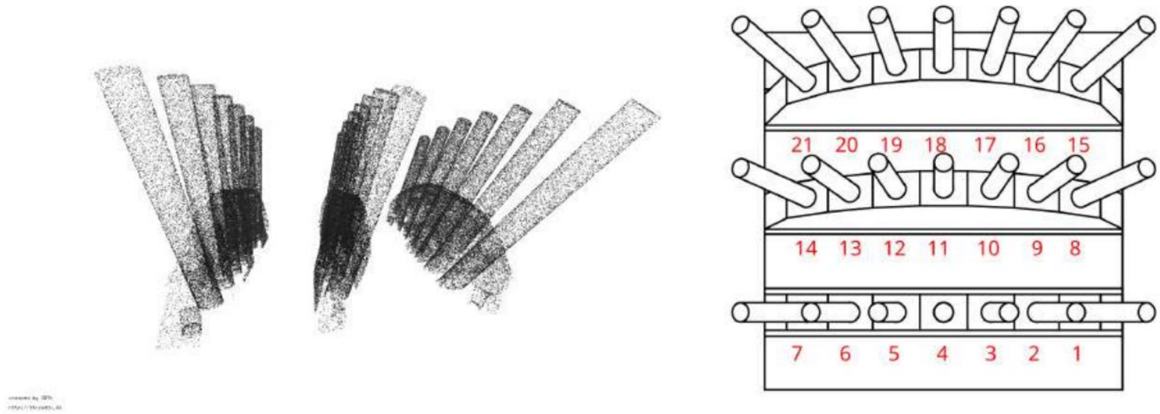
Overall, it can be clearly stated that cylinder detection is successful on simulated data. The errors in circle detection are very small and have a very low spread. Out of 16,000 cylinders, a total of 15,999 cylinders were detected without significant errors.

The functionality of circle detection has thus been verified. For this reason, cylinder detection will be evaluated in the following experiments using both simulation data and real-world data.

cylinder [Radius, height]	Optimal position [u,v][mm]	mean [u,v][mm]	Medium Distance error [mm]	Standard deviation [mm]
Cyl. 1 [10 mm, 25 mm]	[-45.0 -45.0]	[-44.99979 -45.00000]	0.01517	[0,00043 0.00010]
Cyl. 2 [10 mm, 20 mm]	[-15.0 -45.0]	[-14.99989 -45.00029]	0.01620	[0,00022 0.00041]
Cyl. 3 [10 mm, 15 mm]	[15.0 -45.0]	[15,00021 -44.99990]	0.01625	[0.00041 0.00021]
Cyl. 4 [10 mm, 10 mm]	[45.0 -45.0]	[44,99971 -45.00000]	0.01584	[0.00041 0.00002]
Cyl. 5 [7.5 mm, 25 mm]	[-45.0 -15.0]	[-45,00088 -15.00059]	0.05733	[0.00123 0.00082]
Cyl. 6 [7.5 mm, 20 mm]	[-15.0 -15.0]	[-14.99980 -15.00029]	0.01892	[0,00043 0.00042]
Cyl. 7 [7.5 mm, 15 mm]	[15.0 -15.0]	[14,99942 -15.00029]	0.03536	[0.00082 0.00042]
Cyl. 8 [7.5 mm, 10 mm]	[45.0 -15.0]	[44,99971 -14.99990]	0.01622	[0.00041 0.00020]
Cyl. 9 [5.0 mm, 25 mm]	[-45.0 15.0]	[-45,00040 14.99952]	0.03736	[0.00028 0.00101]
Cyl. 10 [5.0 mm, 20 mm]	[-15.0 15.0]	[-14.99950 14.99931]	0.03171	[0.00005 0.00061]
Cyl. 11 [5.0 mm, 15 mm]	[15.0 15.0]	[15,00069 15,00030]	0.03340	[0,00063 0.00043]
Cyl. 12 [5.0 mm, 10 mm]	[45.0 15.0]	[44,99971 15,00029]	0.02230	[0.00041 0.00041]
Cyl. 13 [2.5 mm, 25 mm]	[-45.0 45.0]	[-45,00050 45,00019]	0.02131	[0.00002 0.00061]
Cyl. 14 [2.5 mm, 20 mm]	[-15.0 45.0]	[-14.99900 45,00040]	0.03408	[0.00002 0.00020]
Cyl. 15 [2.5 mm, 15 mm]	[15.0 45.0]	[15,00113 45,00098]	0.05274	[0.00122 0.00102]
Cyl. 16 [2.5 mm, 10 mm]	[45.0 45.0]	[44,99971 45,00050]	0.01984	[0.00041 0.00004]

Table 5.5: Mean error and standard deviation of cylinder positions

5.2 Experiment and evaluation of cylinder detection



(a) Simulated point cloud (b) Representation of cylinder numbering

Figure 5.8: Representation of model 2 for evaluating cylinder detection

In the previous section 5.1.2, the basic functionality of circle detection was tested with a detected cylinder axis. Due to the combined step, cylinder detection was thus tested in the simplest case of a single existing axis. In the following sections, it will be validated that cylinders can also be detected with different axes, even if several plane points are present.

For this purpose, a model with a total of 21 different cylinder axes and a radius of 4 mm was created. The 21 cylinders are evenly distributed across three rows. Each row of cylinders has a uniform angle of 90°, 75° , or 60° to the model plane. The angular difference between two adjacent cylinders in a row is always 10° . Figure 6.2 in the appendix illustrates the model and its angles.

For cylinder detection, the model was sampled and the large planes, such as the model plate, were removed from the point cloud. However, to test the robustness of the detection, some plane points were left in the point cloud. The left-hand figure in section 5.8 shows the simulated point cloud; the side view makes it easy to identify the three rows of cylinders. In the right-hand figure, numbers are assigned to the individual cylinders in red. These numbers will be used for the evaluation in the following sections. Cylinders 1-7 are in row 1, cylinders 8-14 are in row 2, and the remaining cylinders are in row 3. In the following sections, cylinder detection will be tested and evaluated on this point cloud.

5.2.1 Generation of test data and distribution of detected cylinders

Figure 5.9: Representation of the found cylinders (yellow) in a simulated point cloud

As with the validation of circle detection, a sampled 3D point cloud is first generated using the 3D model. After removing the ground planes, the point cloud consists of slightly more than 160,000 points. As before, the cylinder detection is evaluated based on a total of 1,000 detections due to its randomization. This allows for a better assessment of the functionality and robustness of the cylinder detection. Therefore, for the test dataset, each cylinder must be found 1,000 times, resulting in a total of 21,000 cylinders being detected across all runs.

A total of 21,580 cylinders were found, of which 685 were false detections and 20,895 were correctly detected, resulting in 105 false detections. Table 5.6 lists all found cylinders by the number of detections, their mean radius, and their mean axis. If an axis in Hough space had two maxima, one of the maxima was mirrored to the opposite side of Hough space for the purpose of calculating the mean. The total number of all found cylinders corresponds to the 20,895 detections.

From the 20,895 detected cylinders, as well as the 21,000 cylinders actually present, the 105 missing cylinders can be calculated. The reason for this discrepancy lies largely in the cylinder axis detection, whose Hough space is influenced by the plane points. This results in the circle detection frequently passing between 63 and 89 possible cylinder axes to the actual 21 maxima instead of the stated maximum. Most of these axes originate from maxima generated by cylinders, but there are also a few generated by the plane points.

do not correspond to any cylinder axis. These can lead to false detections. Considering the Number of individual detections in 5.6: cylinder 13 was not found a total of 23 times. and several other cylinders, such as 1, 7, 9 and 14, were not detected at least 10 times. The median and simultaneously the maximum number of cylinders detected in one pass The minimum number of correctly detected cylinders in one pass is 21. 19. In Figure 5.9, every 25th detected cylinder is shown for better visualization.

Cylinder [series]	Number of	Medium Radius [mm]	Medium Cylinder Axis
Cyl. 1 [R1]	984	4,017412	[-0,000005 0,867644 0,497098]
Cyl. 2 [R1]	1000	4.019130	[0.000922 0.939864 0.341480]
Cyl. 3 [R1]	999	4.021614	[0.001319 0.984756 0.173802]
Cyl. 4 [R1]	1000	4.021055	[0.000038 0.999972 0.006095]
Cyl. 5 [R1]	1000	4.023453	[-0.001655 -0.984742 0.173892]
Cyl. 6 [R1]	1000	4.018585	[-0.001355 -0.939866 0.341473]
Cyl. 7 [R1]	989	4.017419	[-0.001086 -0.867577 0.497213]
Cyl. 8 [R2]	995	4.012951	[-0.223956 0.836439 0.500157]
Cyl. 9 [R2]	990	4.017824	[-0.243564 0.907487 0.342173]
Cyl. 10 [R2]	992	4.018680	[-0.255035 0.951598 0.171397]
Cyl. 11 [R2]	1000	4.024651	[0.258363 -0.966008 0.007524]
Cyl. 12 [R2]	999	4.022630	[0.253674 -0.951957 0.171428]
Cyl. 13 [R2]	977	4.014294	[0.243128 -0.907709 0.341907]
Cyl. 14 [R2]	987	4.008037	[0.224230 -0.836934 0.499206]
Cyl. 15 [R3]	994	4.012634	[-0.432120 0.74952 0.501454]
Cyl. 16 [R3]	999	4.015868	[-0.470141 0.814086 0.340869]
Cyl. 17 [R3]	1000	4.023035	[-0.491832 0.853793 0.170533]
Cyl. 18 [R3]	1000	4.014919	[0.498872 -0.866663 0.003528]
Cyl. 19 [R3]	999	4.014306	[0.491193 -0.854039 0.171127]
Cyl. 20 [R3]	999	4.012021	[0.469666 -0.814438 0.340685]
Cyl. 21 [R3]	992	4.006345	[0.432675 -0.749493 0.501005]

Table 5.6: Number of detections by each cylinder

Besides the false detections, the large number of false positives is particularly striking. In 1000 runs, 685 incorrect cylinders were detected. These incorrect cylinders are divided as follows: However, they always had the same characteristics. For example, 99 axles were used for the 685 cylinders. detected, but all of these have a z-value around 0.997. At the same time, all false detections have a very small radius in the range of [0.66699; 1.6654]. Thus, there are There are two ways the algorithm can be improved. Firstly, further [details] can be added. Conditions in circle detection, such as a minimum radius, help to identify smaller cylinders.

to avoid outliers, which is particularly important in the case of a falsely detected cylinder axis, as this would result in every projected point being an outlier. Furthermore, improved evaluation of the Gaussian-Hough space can lead to significantly fewer false cylinder axes being detected. For example, additional conditions can be imposed on the neighborhood of the maximum during evaluation, ensuring that only point-like maxima are detected.

In the following sections, the aforementioned difficulties will be revisited and analyzed in more detail, step by step. Particular attention will be paid to the root causes of these difficulties and to potential improvements.

5.2.2 Evaluation of cylinder axis detection on a noise-free point cloud

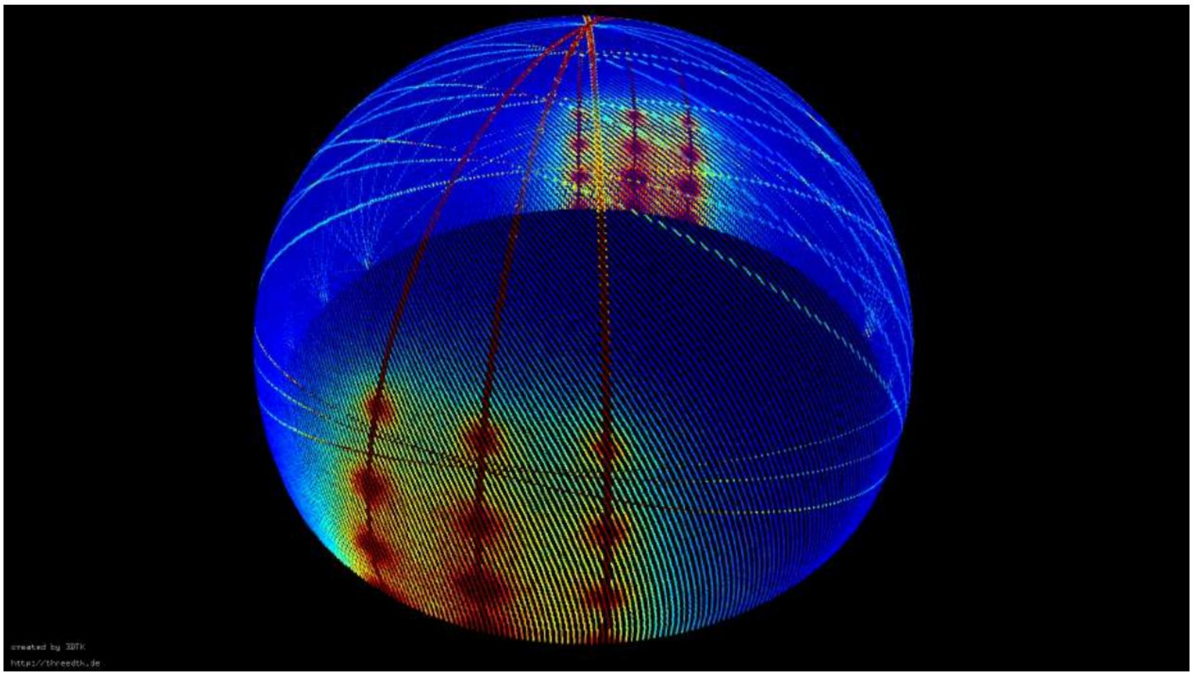


Figure 5.10: Representation of the accumulator for detecting the cylinder axes, where blue represents unincremented cells and red represents cells with maximum accumulation.

The first step in successful cylinder detection is the detection of strong hypotheses for the cylinder axes. The special feature of the model used for cylinder detection is that there are always seven cylinders in a row, all with the same angle to the base plate, and only their inclinations differ from each other within that row. Looking at the resulting accumulator, shown in Figure 5.10, the three rows of seven cylinders each can be clearly identified, with three of the maxima located at the edge of the hemisphere, resulting in a total of 21 + 3 maxima being detected.

In addition to the point-like maxima, circular structures can also be seen.
These arise from the remaining plane points of the support. The points are distributed across six different planes, with two planes always sharing the same normal vector. Thus, the plane points form three circles on the accumulator. For this reason, large planes must be removed from the point cloud before cylinder detection.

Smaller remnants are not a major problem, especially if the cylinder maxima are significantly larger than those of the planes. In this case, however, there are so many plane points that they influence cylinder axis detection. This leads to wider cylinder axis maxima in Hough space and simultaneously to false hypotheses around [0 0 1]. The latter can ultimately lead to the detection of small cylinders in outlier clusters.

One way to avoid false cylinder axis detections is to extend the filter window function used for evaluation. This would require considering not only the nearest neighbors, as is currently the case, but also the wider surroundings to analyze the formation of a maximum. If the maximum exhibits a linear circular structure, it must be removed; if it is point-like, it likely represents the hypothesis of a cylinder axis. Such a filter extension would significantly increase the robustness of cylinder axis detection. Simultaneously, it would minimize the subsequent circle detections required, as the number of detected axes would decrease. In the specific case of this evaluation, the falsely detected cylinders would also be eliminated, since their axis can be described by the intersection of the three circles in Hough space.

cylinder	Number Axles	Middle cylinder axis Target cylinder axis	Mean absolute Error[ÿ]	Standard deviation
Cyl. 1 [R1]	21	[-0.000005 0.867644 0.497098] [0.000000 0.866025 0.500000] 0.008753 [0.001704 0.004579 0.007978] [0.000922		
Cyl. 2 [R1]	23	0.939864 0.341480] [0.000000 0.939693 0.342020] 0.006115 [0.001880 0.002225 0.006114] [0.001319 0.984756		
Cyl. 3 [R1]	17	0.173802] [0.000000 0.984808 0.173648] 0.005957 [0.001886 0.001140 0.006420] [0.000038 0.999972 0.006095]		
Cyl. 4 [R1]	20	[0.000000 1.000000 0.000000] 0.006613 [0.001796 0.000022 0.003976] [-0.001655 -0.984742 0.173892]		
Cyl. 5 [R1]	20	[-0.000000 -0.984808 0.173648] 0.005793 [0.001602 0.001103 0.006208] [-0.001355 -0.939866 0.341473]		
Cyl. 6 [R1]	23	[-0.000000 -0.939693 0.342020] 0.006146 [0.001558 0.002259 0.006211] [-0.001086 -0.867577 0.497213]		
Cyl. 7 [R1]	20	[-0.000000 -0.866025 0.500000] 0.008828 [0.001419 0.004606 0.008020] [-0.223956 0.836439 0.500157] [-0.224144		
Cyl. 8 [R2]	21	0.836516 0.500000] 0.005919 [0.001964 0.003524 0.006385] [-0.243564 0.907487 0.342173] [-0.243210		
Cyl. 9 [R2]	23	0.907673 0.342020] 0.006244 [0.001760 0.002501 0.007252]		

Continued on the next page

Table 5.7 – *Continued from previous page*

cylinder	Number Axles	Middle cylinder axis Should cylinder axis	Medium absolute Error[ÿ]	Standard deviation
Cyl. 10 [R2]	25	[-0.255035 0.951598 0.171397] [-0.254887 0.951251 0.173648]	0.005781	[0.001722 0.001024 0.006054]
Cyl. 11 [R2]	20	[0.258363 -0.966008 0.007524] [0.258819 -0.965926 -0.000000]	0.007952	[0.002052 0.000563 0.003977]
Cyl. 12 [R2]	22	[0.253674 -0.951957 0.171428] [0.254887 -0.951251 0.173648]	0.005832	[0.001321 0.001181 0.006016]
Cyl. 13 [R2]	23	[0.243128 -0.907709 0.341907] [0.243210 -0.907673 0.342020]	0.005856	[0.001929 0.002440 0.006648]
Cyl. 14 [R2]	20	[0.224230 -0.836934 0.499206] [0.224144 -0.836516 0.500000]	0.005863	[0.001842 0.003482 0.006299]
Cyl. 15 [R3]	16	[-0.432120 0.749520 0.501454] [-0.433013 0.750000 0.500000]	0.005041	[0.001541 0.002719 0.005160]
Cyl. 16 [R3]	21	[-0.470141 0.814086 0.340869] [-0.469846 0.813798 0.342020]	0.005442	[0.001073 0.002368 0.005796]
Cyl. 17 [R3]	26	[-0.491832 0.853793 0.170533] [-0.492404 0.852869 0.173648]	0.006836	[0.001813 0.001161 0.007289]
Cyl. 18 [R3]	20	[0.498872 -0.866663 0.003528] [0.500000 -0.866025 -0.000000]	0.004209	[0.001586 0.000913 0.002419]
Cyl. 19 [R3]	29	[0.491193 -0.854039 0.171127] [0.492404 -0.852869 0.173648]	0.007186	[0.001430 0.001404 0.007723]
Cyl. 20 [R3]	19	[0.469666 -0.814438 0.340685] [0.469846 -0.813798 0.342020]	0.005324	[0.001972 0.001880 0.005702]
Cyl. 21 [R3]	22	[0.432675 -0.749493 0.501005] [0.433013 -0.750000 0.500000]	0.005553	[0.001873 0.003158 0.005889]

Table 5.7: Mean cylinder axes, errors and standard deviations

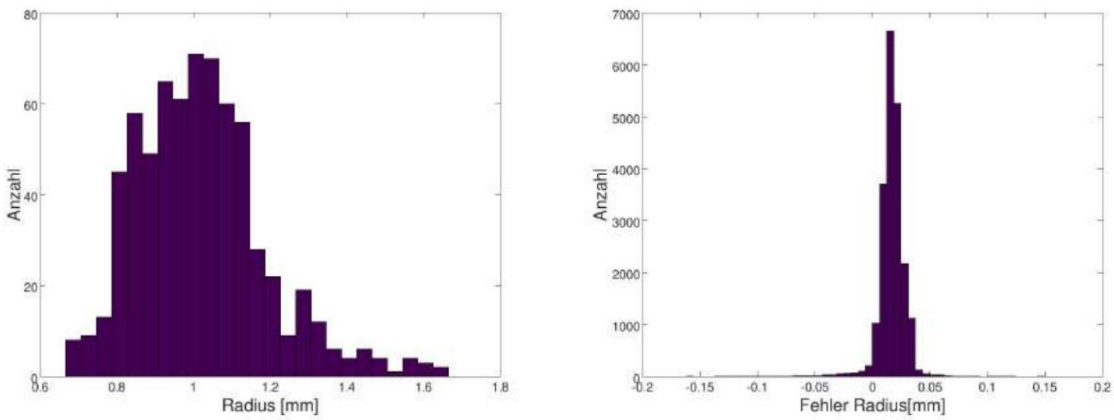
Table 5.7 shows the individual mean axes of the cylinders and their standard deviations. In the case of double maxima, the cylinder axes were aligned to a
The side of the Hough space is mirrored so that the mean is meaningful and the standard deviation can provide information about the spread. In contrast to the evaluation of the
Circle detection requires significantly more axes (approximately 20 per cylinder) across all passes.
found. However, this number increases even further when considering the subsequent analysis.
based on real data, where usually over 50, sometimes even over 100 axes are possible for one cylinder.
are (see Table 5.11). Since there are significantly more level points compared to the first model.
There are 20 possible axes that influence the detected maxima.
With a Hough space size of 636,473 cells, this remains a small number. Furthermore, the standard deviations clearly show that the spread of the axes is very low.

Thus, despite interfering plane points, it can be stated that the cylinder axes were successfully detected with only a small scatter range for cylinder axis detection.

However, due to the plane points, incorrect axes were also detected, which can subsequently lead to false detections. One way to improve this weakness is through an extended evaluation function of the accumulator. Without this extension, a very precise plane detection must be calculated beforehand to achieve a better hypothesis of the cylinder axes, especially with more complex and larger datasets, and the points on the detected planes must be deleted.

5.2.3 Evaluation of circle detection on a noise-free point cloud

In addition to an extended evaluation of the accumulator, it is also possible to minimize the number of incorrect cylinders by imposing further conditions in the circle detection, such as by specifying a minimum radius.



(a) Radii of false detections (b) Radius error of the detected cylinders

Figure 5.11: Detected radii of false detections (left) and errors of correctly detected circles (right)

By examining the radius distribution of the false cylinders shown in the left-hand figure of 5.11, it becomes clear why this is possible. Most of the falsely detected cylinders have a very small radius of approximately 1.0 mm, while the correctly detected cylinders exhibit only a very small error range (compare the right-hand figure of 5.11) and thus all have a radius of approximately 4.0 mm. Adjusting the validation parameters can minimize this problem. For example, increasing the maximum number of positive sectors of a circle required for successful validation can minimize false detections. However, such an adjustment of the validation parameters also increases the number of false detections. Another possibility often arises in reality, since the largest falsely detected circle has a radius of 1.6654 mm and the smallest one of 0.66699 mm.

Thus, for example, by simply introducing a minimum radius of, say,
2.0 mm, minimizing the number of false detections.

However, such a condition not only prevents the detection of very small, false positives
cylinders, but also allows the runtime of circle detection to be improved, especially when
Furthermore, a maximum radius of approximately 8.0 mm is also set. These two conditions
This allows circles whose radii are definitely incorrect to be excluded from the growing accumulator.
to insert, but to delete directly after the calculation. This way, the accumulator doesn't need to be...
The detection area is unnecessarily large, and the circle detection can be faster for potential entries.
test. In the case of this master's thesis, however, the generally valid case is always used without further ado.
Conditions are considered, since adding conditions always affects the time and the result
improved.

cylinder	Radius range [mm]	Medium Radius [mm]	Medium absolute error [mm]	Standard deviation Radius [mm]
Cyl. 1 [R1]	[3.873570, 4.042860]	4.017412	0.021955	0.018160
Cyl. 2 [R1]	[3.920040, 4.052510]	4.019130	0.020352	0.008557
Cyl. 3 [R1]	[3.867790, 4.048450]	4.021614	0.022870	0.011747
Cyl. 4 [R1]	[3.926180, 4.041820]	4.021055	0.022128	0.007909
Cyl. 5 [R1]	[3.880710, 4.101550]	4.023453	0.024644	0.010546
Cyl. 6 [R1]	[3.938140, 4.077900]	4.018585	0.020057	0.010217
Cyl. 7 [R1]	[3.843660, 4.048110]	4.017419	0.021130	0.016531
Cyl. 8 [R2]	[3.898780, 4.053980]	4.012951	0.014993	0.011855
Cyl. 9 [R2]	[3.838110, 4.086050]	4.017824	0.019991	0.014621
Cyl. 10 [R2]	[3.905640, 4.058190]	4.018680	0.019494	0.008811
Cyl. 11 [R2]	[3.931470, 4.045540]	4.024651	0.025484	0.009415
Cyl. 12 [R2]	[3.896700, 4.060400]	4.022630	0.023936	0.011261
Cyl. 13 [R2]	[3,937590, 4,117260]	4.014294	0.015215	0.011965
Cyl. 14 [R2]	[3,893810, 4,024210]	4.008037	0.010173	0.010058
Cyl. 15 [R3]	[3,953360, 4,034800]	4.012634	0.013547	0.006042
Cyl. 16 [R3]	[3,906410, 4,051200]	4.015868	0.016917	0.009431
Cyl. 17 [R3]	[3,878660, 4,149400]	4.023035	0.024821	0.015317
Cyl. 18 [R3]	[3,976850, 4,051830]	4.014919	0.015092	0.004304
Cyl. 19 [R3]	[3,903860, 4,124240]	4.014306	0.015507	0.012861
Cyl. 20 [R3]	[3,918530, 4,032940]	4.012021	0.013412	0.008599
Cyl. 21 [R3]	[3,901680, 4,033120]	4.006345	0.007770	0.006264

Table 5.8: Mean radius, absolute error and standard deviation of the detected radii

Table 5.8 shows the radii of all correctly detected cylinders. The second column clearly shows that the radius ranges are very small. The largest
The radius found is 4.149400mm (cylinder 17) and the smallest is 3.838110mm (cylinder 9).
Thus, the maximum range across all cylinders is just 0.31129 mm. Also, the
Standard deviations clearly show that the spread of the radii is small. For example,
Cylinder 1 has the largest standard deviation at 0.018160 mm. Therefore, the table reflects the
Findings that were already obtained from the visual evaluation of Figure 5.11.
The errors and dispersion of the detected radii are very small, therefore they are
always around the value of 4mm, while all false detections have a significantly smaller radius range.
exhibit.

cylinder	Center axis point [mm]	Distance area to axis [mm]	Medium distance [mm]
Cyl. 1 [R1]	[42.502098 41.822152 65.257740]	[0.000507 0.199956]	0.015829
Cyl. 2 [R1]	[42.499282 44.087813 41.970432]	[0.002480 0.363393]	0.010053
Cyl. 3 [R1]	[42.497363 45.511804 20.584587]	[0.002861 0.240293]	0.008630
Cyl. 4 [R1]	[42.501955 46.229488 0.001379]	[0.002023 0.123379]	0.007813
Cyl. 5 [R1]	[42.499416 45.773216 -20.631825]	[0.001331 0.337462]	0.007752
Cyl. 6 [R1]	[42.501714 43.972568 -41.927370]	[0.001814 0.169967]	0.009328
Cyl. 7 [R1]	[42.500599 41.874985 -65.289221]	[0.001063 0.224524]	0.012072
Cyl. 8 [R2]	[-9.433853 38.929871] 65.778028]	[0.001950 0.181119]	0.009469
Cyl. 9 [R2]	[-10.073417 41.327897 42.373780]	[0.002593 0.267479]	0.014194
Cyl. 10 [R2]	[-13.282702 53.281860 22.686828]	[0.000809 0.150547]	0.023826
Cyl. 11 [R2]	[-10.641500 43.457008 -0.003666]	[0.002319 0.204778]	0.013622
Cyl. 12 [R2]	[-10.476494 42.821427 -20.797641]	[0.002299 0.241416]	0.011043
Cyl. 13 [R2]	[-10.059727 41.293272 -42.362794]	[0.003254 0.240355]	0.027521
Cyl. 14 [R2]	[-9.908514 40.730444 -66.855537]	[0.000307 0.165650]	0.022076
Cyl. 15 [R3]	[-60.923384 36.906518 65.271119]	[0.004304 0.139589]	0.014409
Cyl. 16 [R3]	[-62.406250 39.489644 42.237565]	[0.001984 0.415306]	0.012354
Cyl. 17 [R3]	[-63.747417 41.798975 20.920701]	[0.003202 0.300633]	0.023859
Cyl. 18 [R3]	[-63.407932 41.181783 0.003090]	[0.002213 0.076114]	0.013272
Cyl. 19 [R3]	[-63.276835 40.955831 -20.760079]	[0.003521 0.207553]	0.028102
Cyl. 20 [R3]	[-63.031954 40.547248 -42.690749]	[0.006646 0.171834]	0.021634
Cyl. 21 [R3]	[-61.929947 38.647564 -66.435912]	[0.002536 0.153352]	0.020544

Table 5.9: Error distances of the cylinder axis points

In addition to the radius, the circle detection should also determine a point on the cylinder axis to define the cylinder's 3D position in space. Unlike the initial circle detection evaluation or the later evaluation of the half-cylinders, a comparison with absolute values is very difficult in this case, as each cylinder has a unique axis. Thus, each cylinder has its own projection plane, with the origin depending on the calculated coordinate system.

To still evaluate the points on the cylinder axis, a cylinder center is calculated from the endpoints and starting points of the cylinders. The distance to the cylinder axis, defined by the calculated cylinder center and the absolute cylinder axis, is then calculated. Since the evaluation depends on the calculated mean cylinder center, no statement can be made about a systematic error. However, since no systematic error in the determination of the circle center occurs in either section 5.1.2 or section 5.3, it is assumed that if such an error exists, it is significantly smaller than the random error.

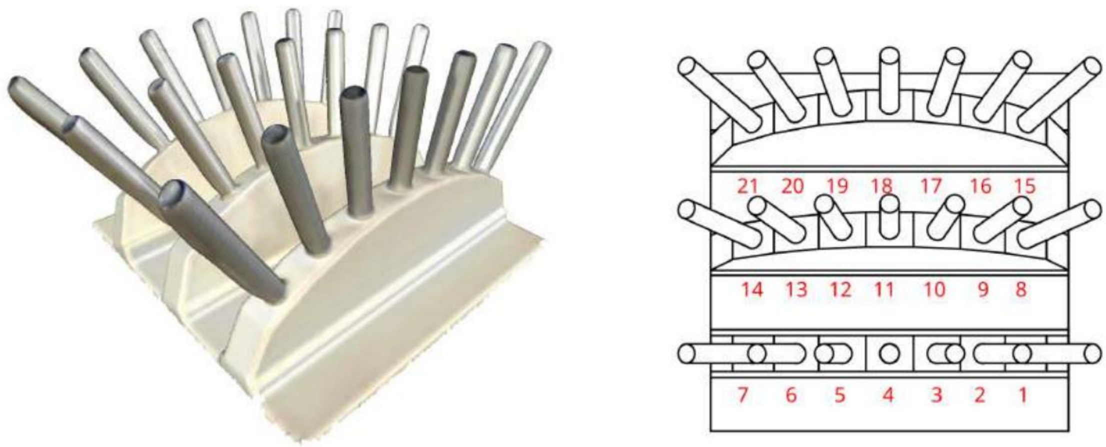
Table 5.9 lists the reference points for the mean axis points. These, along with the cylinder axis, define the line on which all points must lie. In addition to the mean axis point, the table also shows the minimum, maximum, and mean distances to this axis. It is readily apparent that the dispersion of the individual points, as well as the dispersion of the radii, is minimal. However, due to the lack of absolute reference centers, the table provides no information about the absolute deviation, thus preventing the detection of systematic errors. Nevertheless, the distance range and the mean distance provide a rough reference value for random error.

Based on the small distance range to the axis and the radii with very small scatter, it can be concluded that the circle detection is successful despite interfering plane points.
However, the analysis also shows that incorrect cylinder axes can lead to false detections in outliers, which usually have a very small radius. These can be minimized by using more stringent validation parameters, such as a higher number of positive sectors, or by adding conditions.

5.2.4 Evaluation of cylinder detection on a noise-free point cloud

In summary, the algorithm successfully detected cylinders with different axes. Similar cylinder axes can be well represented in a suitably chosen accumulator. However, it should also be clearly noted that a large number of plane points clearly leads to problems. Even in this example, where the large base plate was completely removed and only minor artifacts of the support remain, effects on the Hough space were observed. For instance, in the case of the simulated dataset, significantly more cylinder axes had to be considered for circle detection. Furthermore, some of these axes may be incorrect, which can lead to incorrect circles and thus false detections even in simulation data. Therefore, detection is particularly difficult in a real dataset when plane points are present.

5.2.5 Generation of real scan and categorization of the detected cylinders



(a) Mesh created using the Artec Leo scanner (b) Numbering (red) of the cylinders

Figure 5.12: Mesh and cylinder numbering of the 2nd model

To test the cylinder detection on real-world data, the 3D model (shown in Figure 6.2) was 3D printed, except for the cylinders. Aluminum rods with a diameter of approximately 8.05 mm (measured with calipers) were used in place of the modeled cylinders in the 3D print. The same cylinder numbering system was used for the evaluation as for the simulation data; this numbering is shown again in Figure 5.12 on the right.

The printed model was fully scanned using an Artec Leo scanner. The resulting individual partial scans of the object were then stitched together to form a single image. According to the manufacturer, the Artec Leo has a 3D point accuracy of up to 0.01 mm and a 3D resolution of up to 0.05 mm. However, the point clouds undergo post-processing to help close gaps in the model using interpolation and to correct inaccurate scan sections. [5]

However, post-processing has the major disadvantage that the sharp edges between the cylinder surfaces and the cylinder circles are interpreted as rounding, thus preventing the accuracy specified by the manufacturer from being achieved. In Figure 5.13, two critical areas in the point cloud are outlined in red. Firstly, the tip of the cylinder, which exhibits a rounding due to the post-processing, and secondly, the end of the cylinder, which has a significantly too small radius. Since this post-processing already complicates cylinder detection, most of the plane points on the mount, as well as the model base plate, were removed. Without the plane points, the right-handed scan consists of 424,711 points. As before, not only was detection attempted.

Instead of performing a single test, a total of 1000 runs were used for evaluation. The average execution time for one run was 43.7 seconds (CPU used: AMD Ryzen 5 2600X).

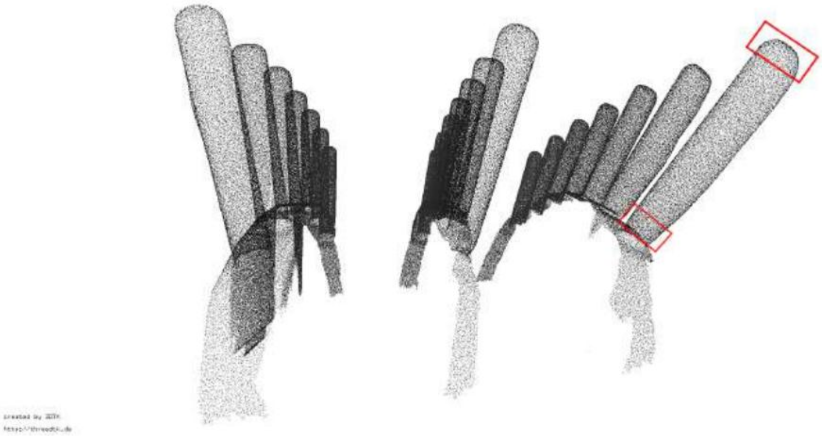


Figure 5.13: Real point cloud of model 2, red: Examples of critical points created by post-processing with the Artec Leo software

In 1000 runs, a total of 21,125 cylinders were found instead of the expected 21,000. Of these, 20,858 cylinders were correctly detected, meaning 142 detections were missed. Cylinder 19 (895) and cylinder 20 (977) were the least frequently detected, while the remaining cylinders all had between 996 and 1000 successful detections. Figure 5.14 shows the middle cylinders of the 1000 detections. It is clearly visible that, due to interpolation, the upper, spherical ends of the cylinders are not detected. Furthermore, some cylinders located at the edge of a row have an obviously too small radius near their mountings and therefore cannot be fully captured.

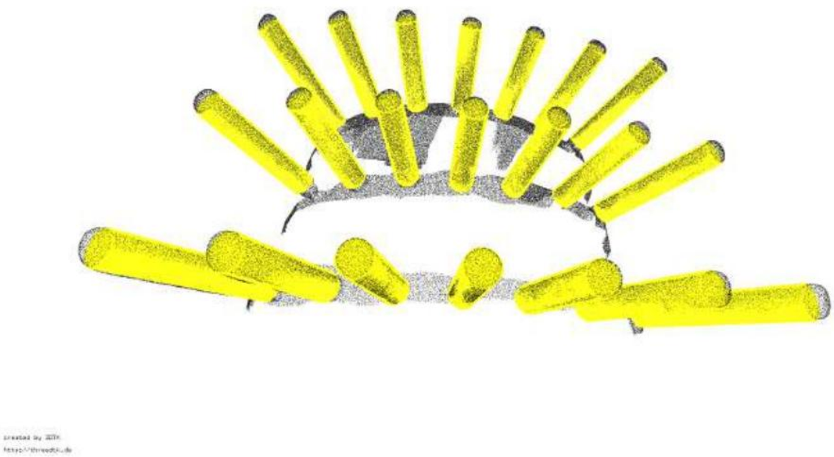


Figure 5.14: Cylinder (yellow) calculated from the mean values of the detected cylinders

267 calculated cylinders are false detections, with this number compared to the simulation.
The number of false detections has decreased due to the smaller number of level points.
as with the simulation data, the radii are small. In total, 224, or approximately 83.9%, possess this characteristic.
All false detections have a radius smaller than 2.5 mm. Possible solutions to this problem.
These issues were already addressed in the evaluation of the simulated data.

For this reason, the errors of the
Cylinder axis detection and circle search are examined in more detail. This will again focus on...
The differences between the simulation data and the real data were discussed.

Cylinder	Number	Average Radius [mm]	Central Axis
Cyl. 1 [R1]	998	4.385837	[0.338382 0.373748 0.863479]
Cyl. 2 [R1]	1000	4.212868	[0.230884 0.259546 0.937531]
Cyl. 3 [R1]	1000	4.091209	[0.122332 0.123496 0.984657]
Cyl. 4 [R1]	1000	3.961372	[0.014676 -0.008713 0.999824]
Cyl. 5 [R1]	1000	3.976495	[-0.117097 -0.151404 0.981366]
Cyl. 6 [R1]	1000	4.057594	[-0.226783 -0.265343 0.936940]
Cyl. 7 [R1]	1000	4.131697	[-0.333609 -0.376526 0.864110]
Cyl. 8 [R2]	999	4.285496	[0.514952 0.221345 0.828044]
Cyl. 9 [R2]	1000	4.132170	[0.418994 0.082546 0.904182]
Cyl. 10 [R2]	1000	4.059775	[0.318332 -0.045989 0.946818]
Cyl. 11 [R2]	999	4.039723	[0.205805 -0.182902 0.961279]
Cyl. 12 [R2]	1000	4.057945	[0.088732 -0.310165 0.946475]
Cyl. 13 [R2]	999	4.107037	[-0.028213 -0.425533 0.904458]
Cyl. 14 [R2]	1000	4.120827	[-0.154003 -0.534709 0.830790]
Cyl. 15 [R3]	998	4.391449	[0.665148 0.077799 0.742590]
Cyl. 16 [R3]	1000	4.353024	[0.584651 -0.057298 0.809168]
Cyl. 17 [R3]	997	4.192846	[0,490671 -0,205716 0,846610]
Cyl. 18 [R3]	996	4.139120	[0.380471 -0.339377 0.860142]
Cyl. 19 [R3]	895	4.146973	[0.259587 -0.468589 0.844249]
Cyl. 20 [R3]	977	4.200681	[0.126788 -0.577034 0.806622]
Cyl. 21 [R3]	1000	4,233473	[-0.000824 -0.670143 0.742200]

Table 5.10: Detected cylinders and their mean radii or cylinder axes

5.2.6 Evaluation of cylinder axis detection on real data

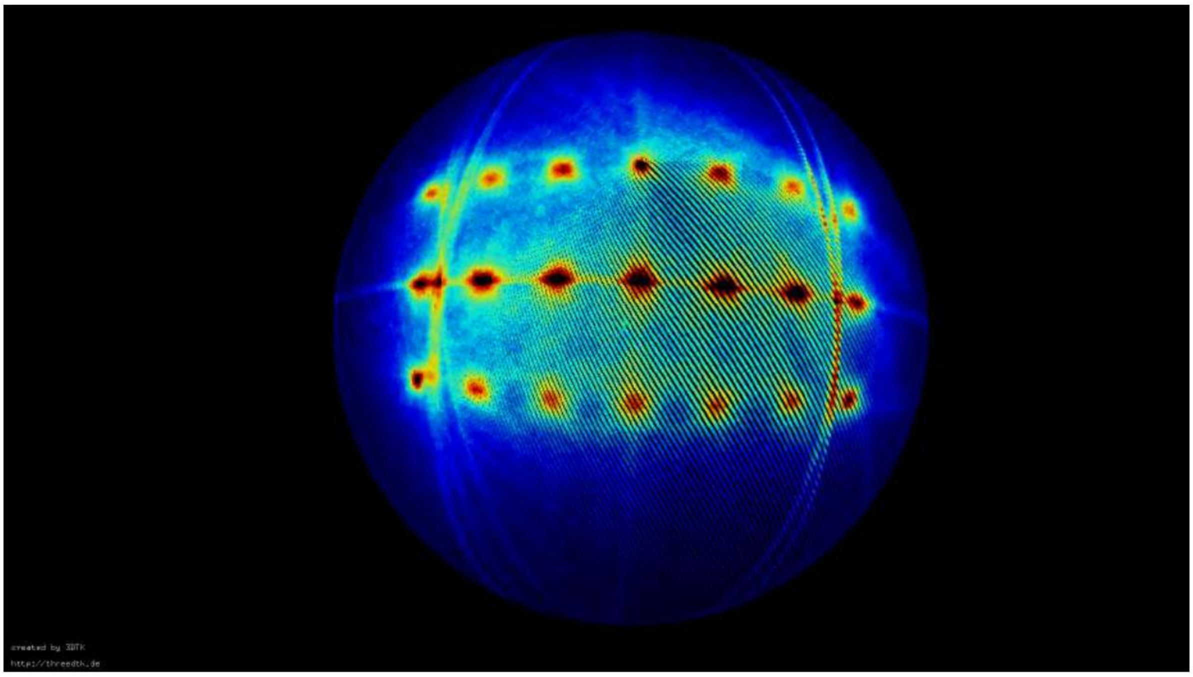


Figure 5.15: Representation of the accumulator for detecting the cylinder axes, where blue does not increment and red describes the maximum number of accumulated cells.

As with the initial cylinder axis evaluation on the simulation data, the accumulator, shown in Figure 5.15, is considered first. In this case, the right-handed coordinate system generated by the laser scanner was used directly, without exporting it as a left-handed coordinate system. Therefore, all maxima are located on the upper side of the accumulator hemisphere. Due to the significantly smaller number of plane points, the maxima of the planes are considerably less pronounced compared to the simulation dataset.

Table 5.11 shows the mean cylinder axes, their standard deviations, and the number of distinct cylinder axes. Compared to the simulated data, most of the detected real cylinders have a significantly larger number of axes. Between 26 and 206 distinct axes were found for the various cylinders. Looking at the standard deviation of the individual values, it is clear that the spread of the detected axes has increased considerably for the real data compared to the simulation data. However, the spread is still within a small range. A direct comparison with absolute values is not possible with real datasets, as only relative references are known in a model.

cylinder	Number Axles	Middle cylinder axis	Standard deviation
Cyl. 1 [R1]	65	[0.338382 0.373748 0.863479]	[0.009994 0.009671 0.004576]
Cyl. 2 [R1]	72	[0.230884 0.259546 0.937531]	[0.014009 0.012168 0.004527]
Cyl. 3 [R1]	68	[0.122332 0.123496 0.984657]	[0.009738 0.011709 0.001748]
Cyl. 4 [R1]	26	[0.014676 -0.008713 0.999824]	[0.005959 0.005114 0.000066]
Cyl. 5 [R1]	66	[-0.117097 -0.151404 0.981366]	[0.010726 0.012786 0.002629]
Cyl. 6 [R1]	75	[-0.226783 -0.265343 0.936940]	[0.012130 0.012012 0.003783]
Cyl. 7 [R1]	64	[-0.333609 -0.376526 0.864110]	[0.011376 0.009990 0.004198]
Cyl. 8 [R2]	159	[0.514952 0.221345 0.828044]	[0.006906 0.009553 0.005933]
Cyl. 9 [R2]	92	[0.418994 0.082546 0.904182]	[0.005380 0.006971 0.002871]
Cyl. 10 [R2]	102	[0.318332 -0.045989 0.946818]	[0.005562 0.007131 0.001781]
Cyl. 11 [R2]	129	[0.205805 -0.182902 0.961279]	[0.007664 0.008546 0.001702]
Cyl. 12 [R2]	125	[0.088732 -0.310165 0.946475]	[0.007410 0.007085 0.002060]
Cyl. 13 [R2]	105	[-0.028213 -0.425533 0.904458]	[0.006704 0.005485 0.002697]
Cyl. 14 [R2]	135	[-0.154003 -0.534709 0.830790]	[0.008337 0.007392 0.005728]
Cyl. 15 [R3]	111	[0.665148 0.077799 0.742590]	[0.004503 0.007016 0.004062]
Cyl. 16 [R3]	135	[0.584651 -0.057298 0.809168]	[0.006733 0.008901 0.004819]
Cyl. 17 [R3]	177	[0.490671 -0.205716 0.846610]	[0.007571 0.009809 0.004645]
Cyl. 18 [R3]	205	[0.380471 -0.339377 0.860142]	[0.009543 0.009606 0.006113]
Cyl. 19 [R3]	204	[0.259587 -0.468589 0.844249]	[0.012762 0.008977 0.006298]
Cyl. 20 [R3]	193	[0.126788 -0.577034 0.806622]	[0.013882 0.009460 0.005982]
Cyl. 21 [R3]	73	[-0.000824 -0.670143 0.742200]	[0.004011 0.004168 0.003760]

Table 5.11: Mean cylinder axis and standard deviation of individual cylinders

For this reason, the angles between the individual axes must be considered. Each The cylinder has at least two reference cylinders that are located in the same position in the two other rows, shifted by an angle of 15° or 30° . Furthermore It is known that the cylinder and its neighbors on a row have an angular difference of 10°. own.

The mean angular difference between the neighbors of a row and between the reference cylinders on the other rows is given in the three tables 5.12, 5.13 and 5.14 for each row. plotted. In addition to the angle difference, the relative absolute mean can also be calculated from the tables. Errors can be read. The error is the relative angular difference between the cylinders. Considering a series, the maximum error is given by 1.7016° (series 1), the minimum 0.42775ÿ(series 3), and the average 0.70012ÿ.

To consider the angular difference between the cylinders of the different series,
The middle row 2 was compared with row 1 and row 3. This has the advantage that the angular difference between the cylinders and their reference cylinders on the other row is always 15°.
This corresponds to the maximum error between the reference cylinders of row 2 and the others.
The series is 2.10785°, the minimum is 0.26532° and the average is 0.98230.

The distribution of the errors of the individual cylinders is shown in Tables 5.12, 5.13 and 5.14.
Upon closer inspection, it is noticeable that the cylinders at the edge of the row have a significantly larger angular difference error to their neighbor in the row, as well as to the two reference cylinders on the other series. The average error angle difference between the cylinder rows, excluding the two outer cylinders, is 0.69712°, with the largest error being just
1.33778° .

cylinder [on R1]	Neighbor on R2 Angle[ÿ]; Error[ÿ]	Neighbor on R3 Angle[ÿ]; Error[ÿ]	Neighbor R1 Angle[ÿ]; Error[ÿ]
Cyl. 1	13.604144; 1.417434;	26.481514; 3.518486	NaN; NaN
cyl. 2	15.293039; 0.882916;	28.807846; 1.235808	9.898373; 0.487555
cyl. 3	14.998413; 0.538410	29.838211; 0.649033	10.479885; 0.688597
cyl. 4	15.040488; 0.289654	29.570354; 0.571482	9.791816; 0.549321
cyl. 5	14.948919; 0.265315	29.701710; 0.461109	11.694049; 1.701605
cyl. 6	14.635565; 0.792643	28.149161; 1.864442	9.037731; 1.020495
cyl. 7	13.817539; 1.186344	26.625785; 3.374215	9.759507; 0.614175

Table 5.12: Relative angles of the cylinders on row 1

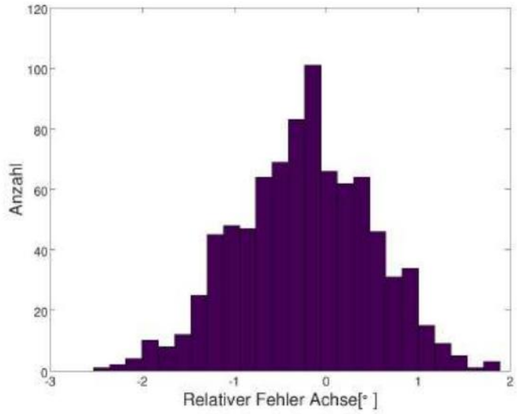
cylinder [on R2]	Neighbor on R1 Angle[ÿ]; Error[ÿ]	Neighbor on R3 Angle[ÿ]; Error[ÿ]	Neighbor R2 1 Angle[ÿ]; Error[ÿ]
Cyl. 8	13.604144; 1.417434;	12.930680; 2.069320;	NaN; NaN
cyl. 9	15.293039; 0.882916;	13.664055; 1.337778;	10.615550; 0.824108
cyl. 10	14.998413; 0.538410	14.809296; 0.471629	9.471513; 0.628227
cyl. 11	15.040488; 0.289654	14.559609; 0.633119	10.412181; 0.589210
cyl. 12	14.948919; 0.265315	14.903793; 0.427439	9.975459; 0.589201
cyl. 13	14.635565; 0.792643	13.669088; 1.332292	9.715666; 0.570653
cyl. 14	13.817539; 1.186344	12.892150; 2.107850	10.728588; 0.813749

Table 5.13: Relative angles of the cylinders on row 2

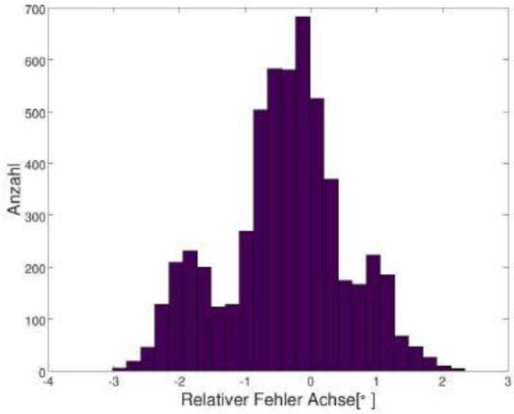
cylinder [on R3]	Neighbor on R2 Angle[ÿ]; Error[ÿ]	Neighbor on R1 Angle[ÿ]; Error[ÿ]	Neighbor R3 1 Angle[ÿ]; Error[ÿ]
Cyl. 15	12.930680; 2.069320	26.481514; 3.518486	NaN; NaN
cyl. 16	13.664055; 1.337778	28.807846; 1.235808	10.014320; 0.427746
cyl. 17	14.809296; 0.471629	29.838211; 0.649033	10.202482; 0.593827
cyl. 18	14.559609; 0.633119	29.570354; 0.571482	9.947050; 0.545626
cyl. 19	14.903793; 0.427439	29.701710; 0.461109	10.250782; 0.569886
cyl. 20	13.669088; 1.332292;	28.149161; 1.864442;	10.191009; 0.748745
cyl. 21	12.892150; 2.107850	26.625785; 3.374215	9.777007; 0.639433

Table 5.14: Relative angles of the cylinders on row 3

This is a significant difference from the previously calculated total average error of 0.98230 and the maximum error of 2.10785°. One reason for this is that in Hough space near the maxima of the outer cylinder axes circles of the remaining These points lie on the plane. They can influence the maxima and prevent accurate detection. significantly complicates matters. This can also be clearly seen in the illustration of accumulator 5.15. become.



(a) Relative error of the cylinder axes on row 1



(b) Relative error between row 1 and row 2

Figure 5.16: Representation of the relative error angles of the cylinder axes

The results from the tables are also shown in Figure 5.16. In the left The figure shows the errors in the relative angles between the cylinders of the first row. The distribution clearly shows that the relative errors have a peak. possessing approximately 0. In the figure on the right, however, the errors of the angle difference between Row 2 and row 1, as well as row 2 and 3, are shown. The small peaks next to them are particularly noticeable. the large maximum at 0, which mainly originates from the two outer cylinders.

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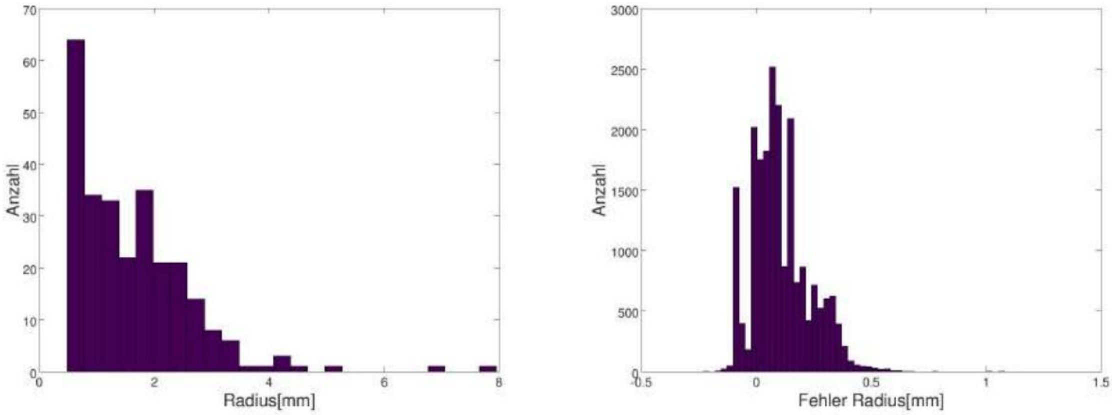
Despite the increasing errors of the edge cylinders, the cylinder axis detection is good enough to enable circle detection based on the point cloud and the detected cylinder axes.

Only by filtering out all plane points would cylinder axis detection benefit even further. However, since planes occur in most real-world cases, future implementations should either employ a very precise plane detection capable of removing even small planes or utilize a more sophisticated evaluation of Hough space.

5.2.7 Evaluation of circle detection on real data



(a) Radii false detections

(b) Radii detected cylinders

Figure 5.17: Comparison of the radii of the false detections (left) with the radii of the correctly detected cylinders (right)

Not only has the scatter of the cylinder axes increased significantly due to the real data, but the calculated radii of the circle detection now also exhibit a considerably larger range of [3.8189 5.1310] mm. This is clearly visible in the right-hand figure of 5.17, which plots the errors of the detected radii. Compared to the clear peak of the simulation data, the radii errors are much more widely scattered here, especially in the positive range.

Table 5.15 shows the detected radii and their absolute mean errors, broken down by cylinder. It is clearly evident that the mean errors of the individual radii are significantly larger than, for example, in the simulation data. It is also noticeable that the error in cylinder axis detection is reflected in the detected radii. For instance, most of the outermost cylinders have a relatively large mean error compared to the remaining cylinders.

If one calculates the average of the absolute mean errors of all outer cylinders, one obtains 0.2097mm, while the absolute mean error of the remaining cylinder radii is 0.0895mm.

This difference clearly shows that the higher the demand on the
The more precise the cylinder detection, the more precise the requirement for the cylinder axis must be.
This is because errors in this are directly reflected in the circle detection. Despite Least-Square
The approach is based on the systematic errors that arise from an incorrect cylinder axis,
It is not possible to completely resolve these issues.

cylinder	Radius range [mm]	mean Radius [mm]	medium absolute error [mm]	Standard deviation Radius [mm]
Cyl. 1 [R1]	[3.992840 4.727300]	4,385837	0.336017	0.091645
Cyl. 2 [R1]	[4.020240 4.331050]	4.212868	0.162933	0.027448
Cyl. 3 [R1]	[3.947030 4.138270]	4.091209	0.041717	0.016627
Cyl. 4 [R1]	[3.921470 3.975950]	3.961372	0.088628	0.004806
Cyl. 5 [R1]	[3.883010 4.119870]	3.976495	0.073762	0.020466
Cyl. 6 [R1]	[3.928220 4.216220]	4.057594	0.020845	0.027529
Cyl. 7 [R1]	[3.818860 4.251030]	4.131697	0.089126	0.051653
Cyl. 8 [R2]	[4.101310 4.460480]	4.285496	0.235496	0.038874
Cyl. 9 [R2]	[3.968260 4.179010]	4.132170	0.082712	0.015291
Cyl. 10 [R2]	[3.981010 4.089620]	4.059775	0.010301	0.005675
Cyl. 11 [R2]	[3.894860 4.097630]	4.039723	0.011510	0.013063
Cyl. 12 [R2]	[3.953280 4.142590]	4.057945	0.010726	0.013018
Cyl. 13 [R2]	[4.000650 4.151010]	4.107037	0.057738	0.014101
Cyl. 14 [R2]	[3.895330 4.235960]	4.120827	0.072855	0.026004
Cyl. 15 [R3]	[4.189280 5.131040]	4.391449	0.341449	0.078991
Cyl. 16 [R3]	[4.120490 4.548210]	4.353024	0.303024	0.054069
Cyl. 17 [R3]	[4.072990 4.290110]	4.192846	0.142846	0.014754
Cyl. 18 [R3]	[3.955020 4.298450]	4.139120	0.089323	0.016568
Cyl. 19 [R3]	[3.939050 4.299490]	4.146973	0.097226	0.023248
Cyl. 20 [R3]	[4.019010 4.406260]	4.200681	0.150785	0.032162
Cyl. 21 [R3]	[4.016400 4.468200]	4.233473	0.183580	0.044864

Table 5.15: Representation of the detected radii by their mean, absolute error and the Standard deviation

Based on the real data, a comparison with absolute positions is not possible, even to
There is no exact information on the relative positions of the cylinder centers, because the height
The cylinder, and therefore the calculated center point of the cylinder, varies greatly (compare
(See values in Table 5.16 for details). This is partly due to the post-processing of the scans, which is necessary for this purpose.
This leads to the fact that, for example, due to interpolation between the lateral surface and the support, as well as
between the lateral surface and the circular area, the radius is of different sizes. If one considers the average lengths,

Thus, the maximum distance between the 21 lengths is approximately 14.2 mm. The range of variation The difference in cylinder lengths is therefore clearly too large for a relative comparison of the individual cylinders to be valid. Points are possible.

Therefore, only the middle points on the axis, the length of the cylinders, are shown in Table 5.16. and the standard deviations of the cylinder axis points are shown. Due to the The standard deviation again highlights the outer cylinders, as these have a significantly larger standard deviation than, for example, the middle cylinders. This also indicates once again what This was actually already well demonstrated by the radius: the greater the error of the cylinder axis, The worse or less accurately the projected circle can be detected, which is why the dispersion of the center point also increases.

cylinder	Cylinder height [mm] Average	Point on axis [mm]	Standard deviation the cylinder axis points [mm]
Cyl. 1 [R1] 51,089 cyl.		[40,840 88,016 98,790]	[0.231 0.475 0.225]
2 [R1] 43.046 cyl. 3		[24,939 70,038 105,560]	[0.182 0.668 0.171]
[R1] 39,419 Cyl. 4 [R1]		[9,265 52,074 109,753]	[0.125 0.631 0.118]
37.505 cyl. 5 [R1]		[-6,522 34,302 110,915]	[0.089 0.488 0.096]
38,560 cyl. 6 [R1]		[-22,024 16,475 109,700]	[0.358 1.000 0.368]
41,937 cyl. 7 [R1]		[-37,869 -1,194 106,163]	[0.452 0.921 0.451]
48,068 cyl. 8 [R2]		[-54,148 -19,387 100,295]	[0.502 0.881 0.496]
45,007 Cyl. 9 [R2]		[83,254 52,340 96,487]	[0.059 0.200 0.059]
40.385 Cyl. 10 [R2]		[67,551 32,558 101,807]	[0.030 0.157 0.030]
36,860 cyl. 11 [R2]		[52,348 13,818 105,421]	[0.049 0.157 0.050]
38.161 cyl. 12 [R2]		[36,531 -4,027 105,419]	[0.091 0.237 0.095]
38,326 cyl. 13 [R2]		[20,696 -21,665 104,688]	[0.112 0.252 0.118]
40.193 cyl. 14 [R2]		[4,100 -39,088 101,689]	[0.136 0.269 0.140]
46,969 cyl. 15 [R3]		[-13,243 -55,975 95,543]	[0.259 0.460 0.269]
48,551 Cyl. 16 [R3]		[122,938 15,361 92,132]	[0,020 0,022 0,020]
41.242 cyl. 17 [R3]		[109,319 -4,872 97,904]	[0.011 0.023 0.011]
39,279 cyl. 18 [R3]		[94,452 -24,126 100,820]	[0.043 0.094 0.041]
37.627 cyl. 19 [R3]		[79,115 -42,295 101,846]	[0.104 0.274 0.100]
38,726 cyl. 20 [R3]		[62,839 -59,469 100,585]	[0.248 0.503 0.239]
42,684 cyl. 21 [R3]		[45,562 -75,756 96,944]	[0.295 0.482 0.280]
46,802		[27,258 -91,877 91,835]	[0.204 0.341 0.205]

Table 5.16: Spread of the cylinder axis centers

5.2.8 Evaluation of cylinder detection on real data

The cylinder detection approach presented in this master's thesis makes it possible to find cylinders even in real-world scans. Detection is possible without restrictions on the five cylinder parameters; however, for accurate detection, the point cloud must meet one condition: it must not contain a large number of plane points. The more plane points there are in the point cloud, the more difficult it becomes to evaluate the Hough space due to the resulting circles. Plane points can therefore not only complicate cylinder detection but also render it impossible if a plane votes higher than the maxima of the individual cylinders. The approach implemented in this thesis is thus only conditionally suitable for direct data analysis without prior plane detection.

5.3 Experiment and evaluation of the detection of half cylinders

Section 5.2 demonstrated the basic functionality of the algorithm using simulation and real-world data from a model with 21 full cylinders. However, the goal of this work is not only to detect whole cylinders, but also partial cylinders, and in the worst case, half cylinders.
This is important because in real-world environments, part of the cylinder may be obscured by an object, such as a wall, or large shadows may cover one side. Furthermore, a scan will never capture an entire cylinder; multiple cylinders are always required. Therefore, this analysis will demonstrate, using horizontally positioned half-cylinders, that the algorithm can also detect partial cylinders.

To test the detection of partial cylinders, in this case half-cylinders, a model with six full cylinders and 16 half-cylinders was created. Figure 6.3 shows the model with the individual cylinders and their lengths. The 16 half-cylinders are arranged horizontally around the six upright full cylinders. All horizontal cylinders have a length of 30 mm. Five of the half-cylinders have a radius of 7.5 mm, while the remaining horizontal and upright cylinders have a radius of 5 mm. The angle between the axes of two adjacent horizontal cylinders with a radius of 5 mm is 18°, and 30° for cylinders with a radius of 7.5 mm. For clarity, these angles are also shown in Figure 6.3.

The following section determines the detection accuracy and error susceptibility of the algorithm for detecting half cylinders on simulation data. Subsequently, the function will be tested for stability using real data. This is important because outliers, noise, and shadows can significantly influence the algorithm's results, especially in the case of half cylinders.

5.3.1 Generation of test data and distribution of detected cylinders

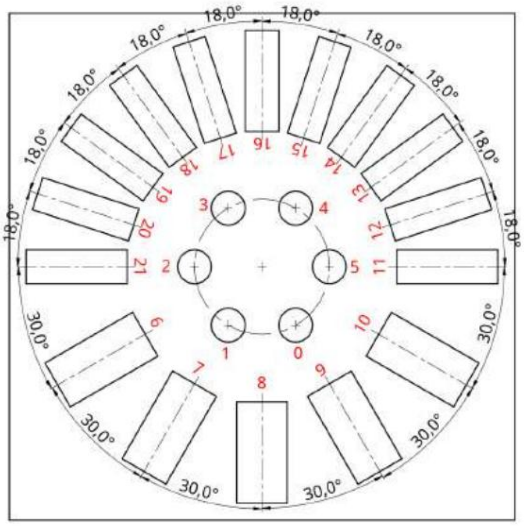


Figure 5.18: Cylinder numbering (red) of the model 3

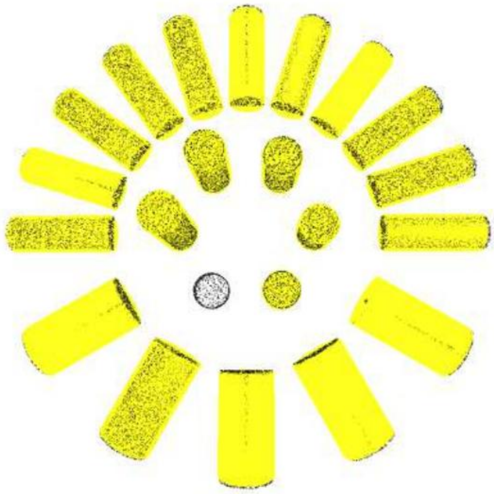


Figure 5.19: Representation of detected cylinders (yellow) in point cloud

For the evaluation of half-cylinder detection on simulation data, the model was sampled with slightly more than 50,000 points. Due to the randomized procedure, as in previous experiments, not a single measurement, but 1,000 cylinder detections were performed on this point cloud. The following section first categorizes the detected cylinders as missing, incorrectly detected, and correctly detected. Subsequently, in section 5.3.2, the first sequential step of cylinder detection is examined in more detail. The evaluation of the radius and the point on the cylinder axis is presented in the following section (5.3.3).

For the evaluation, the cylinder numbering shown in section 5.18 is always used.

Since the model consists of 22 cylinders, of which 6 are full cylinders and the remaining 16 are horizontal cylinders Since half-cylinders are involved, a total of 22,000 cylinders must be detected in 1,000 runs. However, only 20691 cylinders were found, with the best run containing 21 correct cylinders. detected cylinders were found, and the worst was only 19. There were no false detections, therefore... 20,691 out of 220,000 cylinders were correctly detected and 1,309 cylinders were not detected. On average 20.7 cylinders were correctly detected.

cylinder [Radius[mm]; Height [mm]	Whole/Half cylinder	Number	Medium radius [mm]	Middle cylinder axis
Cyl. 0 [5; 5]	whole cyl.	0	-	-
Cyl. 1 [5; 10]	entire cyl.	993	4,999	[-0.000785 0.000000 1.000000]
Cyl. 2 [5; 15]	entire cyl.	1000	4,999	[-0.000785 0.000000 1.000000]
Cyl. 3 [5; 20]	entire cyl.	1000	4,999	[-0.000785 0.000000 1.000000]
Cyl. 4 [5; 25]	entire cyl.	1000	4,999	[-0.000785 0.000000 1.000000]
Cyl. 5 [5; 30]	entire cyl.	1000	4,999	[-0.000785 0.000000 1.000000]
Cyl. 6 [7.5; 30]	half cyl.	1000	7,498	[0.865826 -0.500338 0.000785]
Cyl. 7 [7.5; 30]	half cyl.	1000	7,498	[0.499590 -0.866256 0.000785]
Cyl. 8 [7.5; 30]	half cyl.	1000	7,504	[0.000460 -0.999995 0.000785]
Cyl. 9 [7.5; 30]	half cyl.	1000	7,500	[-0.500316 -0.865838 0.000785]
Cyl. 10 [7.5; 30]	half cyl.	1000	7,497	[-0.865800 -0.500382 0.000785]
Cyl. 11 [5; 30]	half cyl.	924	5,002	[0.999997 0.001097 0.000785]
Cyl. 12 [5; 30]	half cyl.	1000	4,999	[0.951275 0.308333 0.000785]
Cyl. 13 [5; 30]	half cyl.	1000	4,998	[0.809930 0.586518 0.000785]
Cyl. 14 [5; 30]	half cyl.	1000	4,998	[0.588398 0.808564 0.000785]
Cyl. 15 [5; 30]	half cyl.	1000	4,998	[0.308531 0.951209 0.000785]
Cyl. 16 [5; 30]	half cyl.	1000	4,997	[-0.000460 0.999995 0.000785]
Cyl. 17 [5; 30]	half cyl.	999	4,957	[-0.309401 0.950926 0.000785]
Cyl. 18 [5; 30]	half cyl.	1000	4,953	[-0.589411 0.807827 0.000785]
Cyl. 19 [5; 30]	half cyl.	1000	4,998	[-0.809605 0.586967 0.000785]
Cyl. 20 [5; 30]	half cyl.	1000	4,998	[-0.951346 0.308112 0.000785]
Cyl. 21 [5; 30]	half cyl.	775	5,000	[-1.000000 -0.000097 0.000785]

Table 5.17: Number of detections by each cylinder

Since this is significantly smaller than the expected 22, the number of Detections broken down by individual cylinder. It is clearly visible that the smallest A solid cylinder with a radius of 5mm and a height of 5mm is not detected. This also applies to optical detection. Examining one of the 1000 cylinder detections, shown in 5.19, it is clearly visible that

The smallest cylinder was not marked. This is due to the small number of lateral surface points, which prevents the circle validation function, with a circle resolution of 0.1 and at least one point in 25% of all circle segments, from validating it as a circle.

One way to detect this smallest cylinder is to adjust the validation parameters of the circle detection. Increasing the step size of the circle resolution results in fewer circle segments, meaning fewer points need to lie on a circle.

For a resolution of 0.2 mm, 12 false detections were observed in a total of 100 trials, resulting in six false positives. The reason for this is the circular arrangement of the upright solid cylinders. With a circular segment resolution that is too low, the six upright cylinders are detected as one large cylinder with a radius of 40 mm. Depending on the application, however, it is also possible to detect cylinders with only a few lateral points if the cylinders are not arranged circularly in space and the number of plane points is very small. Since the approach will also be tested on real-world data in the following analysis, a robust parameter set for circular validation was chosen for this evaluation. This set ensures that one cylinder is not detected, but also that no false detections occur.

Nevertheless, a mean value of 20.7 is still significantly lower than 21. Table 5.17 shows that a total of 17 cylinders were detected 1000 times. The 309 missing detections are distributed among cylinders 1, 11, 17, and 21, with cylinder 17 only failing to detect once and potentially having 7 false detections. Both cylinder detections can be further improved, for example, by increasing the minimum number of RHT passes for circular detection.

This would not be possible for cylinders 11 and 21, which together failed to detect a total of 301 times. The false detections occur in this case because both cylinders have a radius of 5 mm and lie on the same axis, which is why the circle detection cannot find them in a single pass. If any two cylinders z_i, z_k with the same axis w are translated along w, their circles are superimposed on each other when projected onto the plane. In such a case, only one circle center point c_m is calculated for both cylinders. Thus, the endpoints of only one of the two cylinders, e.g., z_i, are calculated. Subsequently, its points are removed from the input point set of the circle detection, and circle detection is performed again on the remaining points.

Ideally, this second circle detection also finds c_m for the remaining cylinder z_k. However, if z_i was not perfectly detected, for example due to an error in its axis, points from z_i may remain in the circle input set. This can lead to z_i being detected again instead of z_k. Since z_i has already been found once, the renewed detection is discarded, and no further circle search is initiated due to the lack of new detections. This bug should be fixed in later implementations. In total, only 8 false detections occur that were not caused by this problem.

In the following sections, the individual deviations of the found cylinders will be analyzed in more detail, and some of the problems already mentioned will be discussed in greater depth.
A serious error was found in the simulated data for a marginal case. To obtain more robust results, this should be fixed in future implementations.

5.3.2 Evaluation of cylinder axis detection on a noise-free point cloud

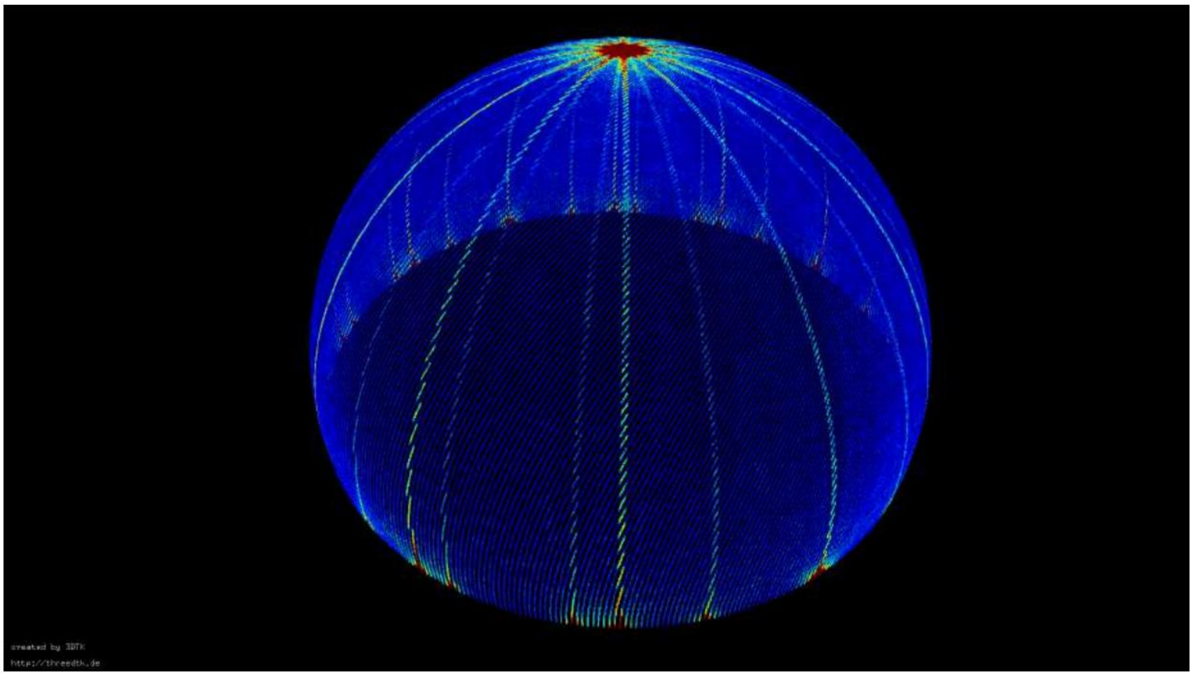


Figure 5.20: Representation of the accumulator for detecting the cylinder axes, where blue represents unincremented cells and red represents cells with maximum accumulation.

After a preliminary evaluation of detected and falsely detected axes, this section will examine cylinder axis detection in more detail. The number of detected axes does not correspond to the number of cylinders, as all upright cylinders share an axis with the value [0 0 1]. The number of axes also differs from the number of maxima in the accumulator. Since the half-cylinders are all perpendicular to the upright solid cylinders, they each have two maxima in Hough space on the opposite side of the hemisphere. For example, the five half-cylinders with a radius of 7.5 mm have 10 possible axes that can be detected. Figure 5.20 illustrates Hough space through the accumulator with its maximum. The figure clearly shows that the maxima of the upright cylinders (red dots on the top of the hemisphere) are significantly larger than the maxima of the horizontal cylinders at the edge of the sphere. This is because a total of 6 cylinders vote for the axis [0 0 1], while usually one half-cylinder votes for two maxima.

The double maxima can be seen not only in the accumulator, but also in the different values of the cylinder axes and must be appropriately mirrored for evaluation, for example for calculating a mean value in Table 5.17.

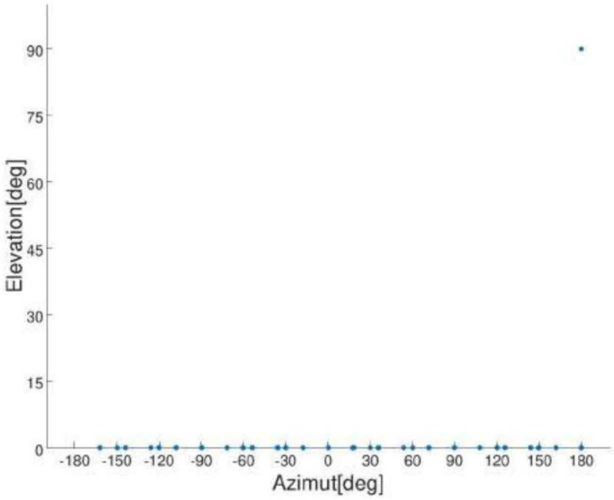
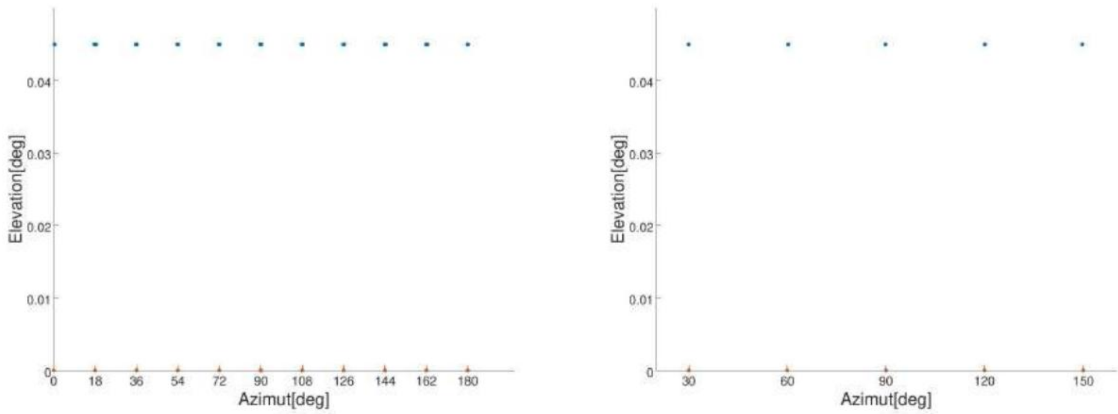


Figure 5.21: Representation of the spherical axes of all cylinders



(a) Axes of the horizontal cylinders with radius 5mm (b) Axes of the horizontal cylinders with radius 7.5mm (blue: Actual, red: Target) (blue: Actual, red: Target)

Figure 5.22: Representation of the cylinder axes in spherical coordinates

In section 5.21, the normalized axes are represented in spherical coordinates to allow direct observation not only of the distribution of the axes but also of the angles between them. While the axis of the upright cylinders [-7.8540e-04 9.6183e-20 1.0000e+00] lies at an azimuth of 180° and elevation of 90° , the axes of the horizontal half-cylinders have an elevation of approximately 0°. However, in this range, 28 unique axes are plotted instead of the 14, with cylinders 11 and 21, as well as 8 and 16, sharing the same axis. This is again due to the double maxima of Hough space.

If one considers the axes of the horizontal cylinders with a radius of 7.5 mm, or analogously all Half of a horizontal cylinder with a radius of 5 mm can be measured individually, so the angle difference of The 30° and 18° angles between the individual axes are clearly visible, as is the angle difference. in the plane corresponds to the angular difference in azimuth (see 5.22). For a clearer presentation, all axes with negative azimuth are shown in Figure 5.22. mirrored and placed on the corresponding opposite axis in the accumulator. No Complete inversion is used because Hough space is represented as a half-sphere and Thus, the third coordinate of the cylinder axis is always positive, since it is on the first row above the equator of the accumulator lies there. Since in this case *ÿÿ = 0.00079* ° , the error is very small. and negligible.

The graphical representation of the axes in Figure 5.22 already clearly shows that the detected Cylinder axes are within the correct value range. However, for a more precise analysis, the Values of the individual cylinder axes are considered. These are listed individually in Table 5.19. It is possible to directly display all detected axes instead of considering the mean value, since there is a minimum of one and a maximum of three axes for each cylinder. They will be detected. Since the same axis was detected for the stationary cylinders 1-5, These are summarized in the table. The horizontal cylinders, however, all have at least [number missing]. Two detected axes that are opposite each other in the accumulator. In addition to the detected The errors of the axes were also plotted. To improve the accuracy of the axes, In order to be able to estimate, in addition to the error-free, perfect axis, the angle between this and the detected object was also calculated, whereby the maximum error of the angle is approximately 0.51°. The difference between the detected and actual axis is therefore small.

The relative angles shown as examples in 5.18 also clearly demonstrate that only a small The error lies on the axes. Considering the angular difference between the cylinder neighbors... In a circle, the difference will never be exactly 30°, but the error is very small. and accurately reflects the absolute errors of the cylinder axes.

cylinder neighbors	Middle cylinder axis 1	Middle cylinder axis 2	Angle [ÿ]
Cyl. 6 - 7	[0.865826 -0.500338 0.000785]	[0.499590 -0.866256 0.000785]	29.954297
cyl. 7 - 8	[0.499590 -0.866256 0.000785]	[0.000460 -0.999995 0.000785]	30.005269
cyl. 8 - 9	[0.000460 -0.999995 0.000785]	[-0.500316 -0.865838 0.000785]	29.947625
cyl. 9-10	[-0.500316 -0.865838 0.000785]	[-0.865800 -0.500382 0.000785]	30.048238

Table 5.18: Angle difference between the cylinder axis neighbors of cylinders 6-10 (target value is 30°)

Overall, axis detection on the simulated data is possible despite the half cylinders. The detected axes have a very small error rate and As already mentioned, cylinder axis detection is not sufficient to account for the many false detections of Cylinders 11 and 21 are responsible. The crucial limiting factor here, unless there are major However, the detection of circles is based on planes within the scene. An evaluation of this takes place in the next section.

Cylinders	Detected Axles Cyl. 1-5	Perfect axis	Error angle[ÿ]
Cyl. 6	[-0.000785 0.000000 1.000000]	[0.000000 0.000000 1.000000]	0.000000
	[-0.863390 0.504537 0.000785]; [0.866549 -0.499091 0.000785]	[0.866019 -0.500011 0]	0.300572; 0.083651
Cyl. 7	[-0.501816 0.864974 0.000785]; [0.496361 -0.868116 0.000785]	[0.500011 -0.866019 0]	0.128522; 0.243678
Cyl. 8	[0.003148 -0.999995 0.000785]; [0.003148 0.999995 0.000785]	[0.000000 -1.000000 0]	0.181185; 0.181185
Cyl. 9	[-0.501816 -0.864974 0.000785]; [0.496361 0.868116 0.000785]	[-0.500011 -0.866019 0]	0.128522; 0.243678
Cyl. 10	[-0.863390 -0.504537 0.000785]; [0.866549 0.499091 0.000785]	[-0.866019 -0.500011 0]	0.300572; 0.083651
Cyl. 11	[-1.000000 0.000000 0.000785]; [0.999980 0.006296 0.000785]	[1,000000 0,000000 0]	0.000000; 0.362371
Cyl. 12	[-0.950862 -0.309616 0.000785]; [0.948893 0.315596 0.000785]; [0.952792 0.303623 0.000785]	[0,951056 0,309018 0]	0.026597; 0.401829; 0.327141
Cyl. 13	[-0.808276 -0.588803 0.000785]; [0.804553 0.593880 0.000785]; [0.811967 0.583703 0.000785]	[0,809028 0,587770 0]	0.091797; 0.437740; 0.289782
Cyl. 14	[-0.586256 -0.810125 0.000785]; [0.591345 0.806418 0.000785]	[0.587770 0.809028 0]	0.126616; 0.260621
Cyl. 15	[-0.306621 -0.951831 0.000785]; [0.300622 0.953743 0.000785]; [0.312607 0.949882 0.000785]	[0,309018 0,951056 0]	0.158545; 0.507382; 0.224168
Cyl. 16	[0.003148 -0.999995 0.000785]; [0.003148 0.999995 0.000785]	[0.000000 1.000000 0]	0.181185; 0.181185
Cyl. 17	[-0.306621 0.951831 0.000785]; [0.312607 -0.949882 0.000785]	[-0.309018 0.951056 0]	0.158545; 0.224168
Cyl. 18	[-0.586256 0.810125 0.000785]; [0.591345 -0.806418 0.000785]	[-0.587770 0.809028 0]	0.126616; 0.260621
Cyl. 19	[-0.808276 0.588803 0.000785]; [0.811967 -0.583703 0.000785]	[-0.809028 0.587770 0]	0.091797; 0.289782
Cyl. 20	[-0.950862 0.309616 0.000785]; [0.952792 -0.303623 0.000785]	[-0.951056 0.309018 0]	0.026597; 0.327141
Cyl. 21	[-1.000000 0.000000 0.000785]; [0.999980 0.006296 0.000785]	[-1.000000 0.000000 0]	0.000000; 0.362371

Table 5.19: Detected cylinder axes and error angles

5.3.3 Evaluation of circle detection on a noise-free point cloud

The crucial limiting factor in the detection of partial cylinders, specifically half-cylinders, is the circle detection. Its validation parameters must be chosen to be strong enough to avoid circle detections at random points, but at the same time flexible enough to also find semicircles that have significantly fewer points than a full circle.

In the sampled example model, the half-cylinders lying on their side with a length of 30 mm have a significantly larger lateral surface area than, for example, the whole, upright cylinder with a height of 5 mm and a radius of 5 mm. The number of detected cylinders in Table 5.17 shows that this small, upright cylinder is the only one that was not detected. The difficulty with the upright cylinder is that, due to its small size, it has very few points on its lateral surface. It has only 840 points, and the cylinder's circular area is not used for detection; only the points on the lateral surface are considered. This leaves only about 560 points available for detecting the cylinder. Furthermore, due to errors in the spherical normal calculation, these points are not fully utilized for circle detection. Normals near the edge of the circular area can exhibit a large error.

The decisive factor that ultimately leads to the cylinder not being detected is the validation function of the circle detection. For half cylinders, at least 25% of all circle segments must contain a point. The number of circle segments is given by:

$$n_{seg} = \mathrm{ceil}\left(\frac{2 \cdot \pi \cdot r}{b}\right) = \mathrm{ceil}\left(\frac{2 \cdot \pi \cdot 5}{0{,}016}\right) = 1964 . \tag{5.1}$$

Therefore, at least 25% of all 1964 segments, i.e., 491, must contain a point. This is only 69 points fewer than the actual 560 points, which, in the optimal case, would all be projected onto the plane, with all points lying in different circular segments. However, if minor errors occur in the spherical normal calculation, not all lateral points are projected onto the plane, preventing the cylinder's circle from being positively validated. Adjusting the step size of the individual circular segments can help circumvent this problem, but this can lead to an increase in false detections. In the special case of the model described in Figure 6.3, it is possible that instead of the six upright cylinders arranged in a circle, a cylinder with a radius of 40 is detected. Therefore, a reduction in the number of circular segments in the validation function is omitted.

Considering the mean detected radii of the full cylinders compared to that of the half cylinders, as shown in Table 5.20, a small difference in errors and standard deviation can be observed. The mean radius of the horizontal cylinders with a radius of 5 mm is slightly smaller at 4.9905 mm than the detected radius of 4.9992 mm for the full cylinders, resulting in an error that is 0.00870 mm larger. However, this difference is almost negligible compared to the standard deviation, which is 0.019666 mm for the horizontal cylinders with a radius of 5 mm and thus significantly larger than that of the vertical cylinders at 8.0149 e-05 mm. The half cylinders with a radius of 7.5 mm also exhibit a significantly increased

Standard deviation. Nevertheless, the deviation between true and detected radii is
In the case of simulation data, the effect is very small.

cylinder	Radius [mm]	Medium Radius [mm]	Standard deviation [mm]
Entire cylinder center	5	4.9992	8,0149e-05
Half cylinders	7.5	7.4994	0.014368
Half cylinders	5	4.9905	0.019666

Table 5.20: Comparison of mean radii and standard deviation

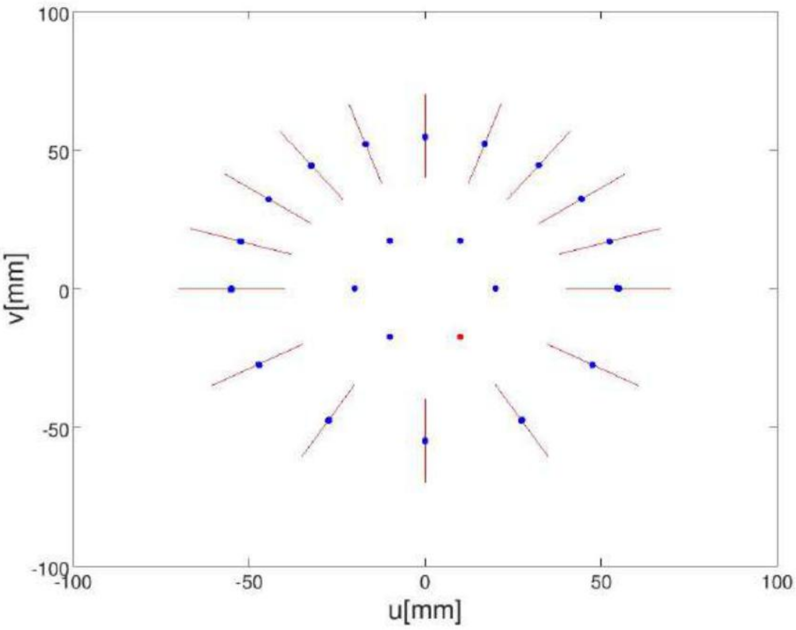


Figure 5.23: Representation of detected cylinder axis centers (blue) and projected cylinder axes (red)

Besides the radii, the positions of the cylinders are also particularly important for circle detection.
Interesting. For this, the detected circle center is converted back into three dimensions.
The coordinate system is transformed to obtain a point on the cylinder axis.
Figure 5.23 shows the points on all cylinder axes projected onto the model plane.
The red lines, or dots, represent the perfect cylinder axes of the individual cylinders. The blue dots correspond to the detected points on the
the found cylinder axis. Since the axes of the standing cylinders are projected onto the plane
If they result in a point, they are obscured by the blue detected cylinder axis centers, with the exception of one point, which represents the smallest, undetected, stationary cylinder.
This represents [something]. For good detection, these must be aligned as precisely as possible with the actual axes.

Visually, all blue dots lie on their respective axes, through which
With a small dispersion, they also usually appear as a single point. Table 5.21 further lists...
firstly, the mean points of the axes, their mean errors, and the length based on them.
of the cylinder.

Cylinder center point on axis	[mm]	Medium Error distance [mm]	distance Standard deviation [mm]	Cylinder length/ Error [mm]
Cyl. 1	[-9,999 -17,320 4,276]	0.000500	0.000000	7,687/-2,313
cyl. 2	[-20,000 0,000 7,637]	0.000000	0.000000	14,267/-0.733
cyl. 3	[-10,001 17,321 9,738]	0.001118	0.000000	17,868/-2,132
cyl. 4	[9,999 17,321 12,668]	0.001118	0.000000	22,978/-2,022
cyl. 5	[20,000 0,000 14,914]	0.000001	0.000008	29,402/-0,598
cyl. 6	[47,618 -27,510 0,002]	0.016817	0.008367	28,704/-1,296
cyl. 7	[27,413 -47,503 0,004]	0.031110	0.011804	28,794/-1,206
cyl. 8	[0,011 -54,877 0,001]	0.053032	0.004220	28,622/-1,378
cyl. 9	[-27,454 -47,566 0,003]	0.061595	0.008217	28,667/-1,333
cyl. 10	[-47,348 -27,334 0,001]	0.016116	0.012676	29,201/-0.799
cyl. 11	[55,055 0,006 -0,001]	0.087294	0.007348	28,576/-1,424
cyl. 12	[52,458 17,045 0.001]	0.008394	0.000769	28,910/-1,090
cyl. 13	[44.534 32.355 0.001]	0.018713	0.005937	28,719/-1,281
cyl. 14	[32.369 44.552 0.003]	0.063396	0.002060	28,447/-1,553
cyl. 15	[16.964 52.205 0.003]	0.037086	0.001924	28,801/-1,199
cyl. 16	[-0.001 54.764 0.003]	0.045609	0.000607	28,418/-1,582
cyl. 17	[-16.906 52.137 0.119]	0.035108	0.006759	28,600/-1,400
cyl. 18	[-32.320 44.438 0.126]	0.063229	0.012007	28,439/-1,561
cyl. 19	[-44.394 32.255 0.000]	0.019965	0.001930	28,988/-1,012
cyl. 20	[-52.297 16.992 0.004]	0.078305	0.002407	28,500/-1,500
cyl. 21	[-55.079 -0.000 0.040]	0.002329	0.021854	28,499/-1,501

Table 5.21: Mean point on the cylinder axis, mean error distance, and standard deviation

It is clearly evident that the mean distance error, with a maximum value of 0.087294 mm, is very small. The difference between the individual error distances, described by the standard deviation, also has a maximum value of only 0.021854 mm. Due to the chosen approach for calculating the cylinder ends, the cylinder length is always less than the actual value. This calculation uses the input point set from the circle detection, which now only includes points whose normals are considered to be the cylinder axis. Since the lateral points near the cylinder edge usually do not have the same normals as the cylinder axis, they are disregarded for cylinder end detection.

Finally, it should be noted that circle detection works as long as no boundary conditions occur, such as two cylinders with the same axis and radius, separated only by a displacement along their axis. The choice of validation parameters is important, as it minimizes outliers, but at the same time, it also sets strict limits for cylinders with a small number of surface points.

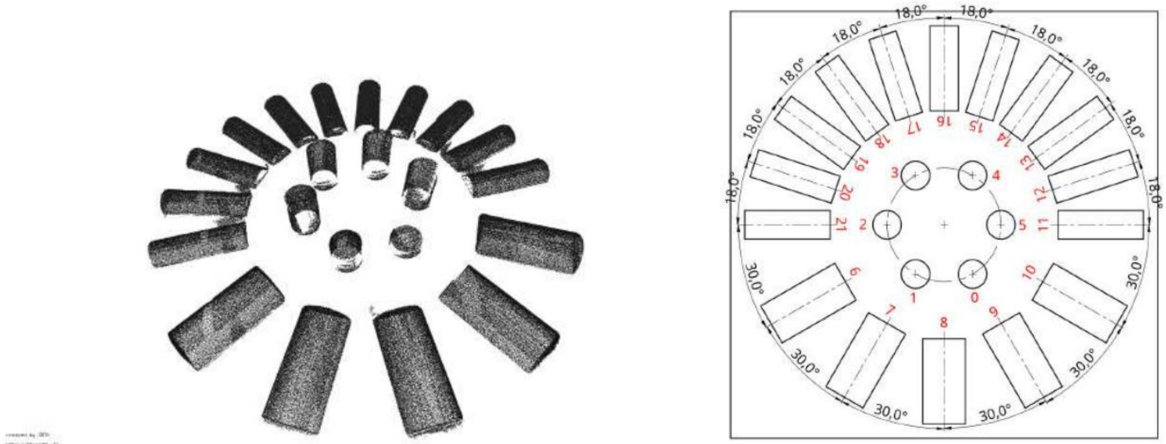
5.3.4 Evaluation of cylinder detection for half cylinders

Although this work seeks a generally applicable solution for cylinder detection, this example clearly illustrates its limitations. For the detection of partial cylinders, it can be stated that the algorithm has two major restrictions.

The first is given by the conditions of the validation parameters of the circle detection.
The standard parameters sometimes need to be adjusted for different point clouds, especially when cylinders have few surface points or many outliers. In some cases, finding an optimal set is very difficult; compare this to the previous problem with upright cylinders. Furthermore, the algorithm should be extended so that multiple cylinders with the same radius on one axis, separated only by a translation along that axis, are also detected without error. However, it should also be noted that the detection of the remaining cylinders is good, and the position, radius, and axis all exhibit only very small errors. The detection of half-cylinders is not a problem on simulated data.

5.3.5 Generation of real scan and categorization of the detected cylinders

To subsequently test the detection of partial cylinders on real-world data, the model (shown in Figure 6.3) was 3D printed. The model was then scanned using a Micro-Epsilon scanControl 2900-25 [4]. To create a 3D point cloud from the 2D data, a KUKA KR 16 was used to scan the model line by line from two different directions, following the trajectory of the industrial manipulator, resulting in two superimposed point clouds. The 3DTK toolkit [1] was used to match these. In addition to plane detection, this toolkit also allows for matching two point clouds using ICP and exporting them as a single left-handed point cloud, as shown in Figure 5.24.



(a) Point cloud of the model

(b) Cylinder numbering (red) of the Models

Figure 5.24: Point cloud and cylinder numbering of model 3

Unlike the scan of section 5.2, no further post-processing took place. Thus There are large gaps in this area, for example due to shadows, which can be clearly seen in Figure 5.24. which can be recognized in the standing cylinders. At the same time, however, the edges of the actual The cylinder was retained, so that the cylindrical circular surface is not spherical but flat. is depicted.

Since section 5.2 explains the effects of planes on cylinder axis detection, the base plate was almost completely removed from the scan. In total The point cloud consists of 238,788 points. Again, instead of a... Detection per 1000 detections considered, with the mean duration of a detection being 37.5s The CPU used is an AMD Ryzen 5 2600X. For the following paragraphs, the cylinder numbering shown in the left-hand figure of 5.24 will be used.

In 1000 runs, instead of the expected 22000 or 21000, the smallest Full cylinders were not detected even on simulated data, a total of 20796. Cylinders found. In contrast to section 5.2, a direct division of the cylinders into Determining correct and incorrect, or missing, cylinders is very difficult, since Many false detections have axes similar to those of the half-cylinders. Since no absolute values exist due to the real data, the cylinders are identified by their cylinder axis difference. The relative target angle to their neighbors is best filtered out. Their medians determine the mean starting and ending points of the cylinder axes. and used as a reference for dividing the cylinders. Then all cylinders are their Start and end points that are closer than *tdis* to their respective reference points are filtered out. For a maximum distance of 5.0 mm, a total of 19,376 cylinders were filtered out of the 20,796 cylinders. This is slightly more than 93% of all cylinders. The cylinders found are referred to below as detected. Thus, a total of 92.3% of all cylinders out of 21,000 were found.

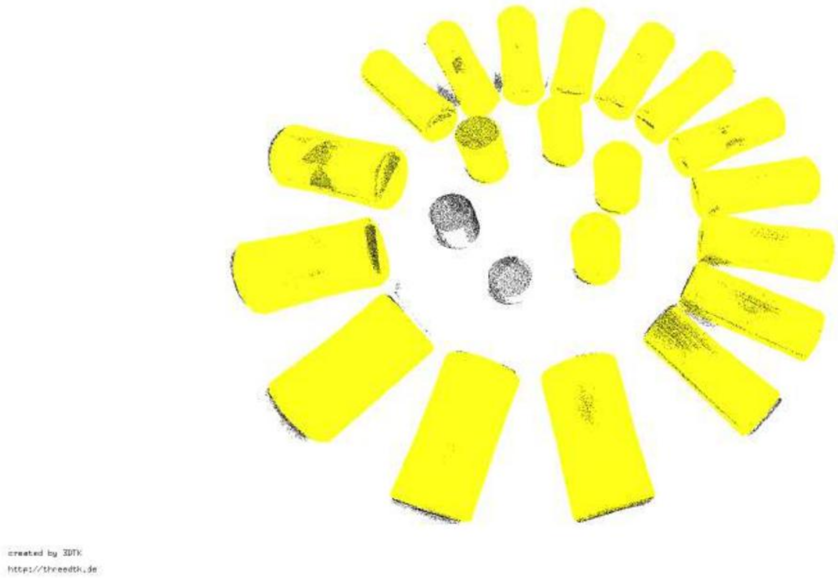


Figure 5.25: Successful detections (yellow) in a real point cloud

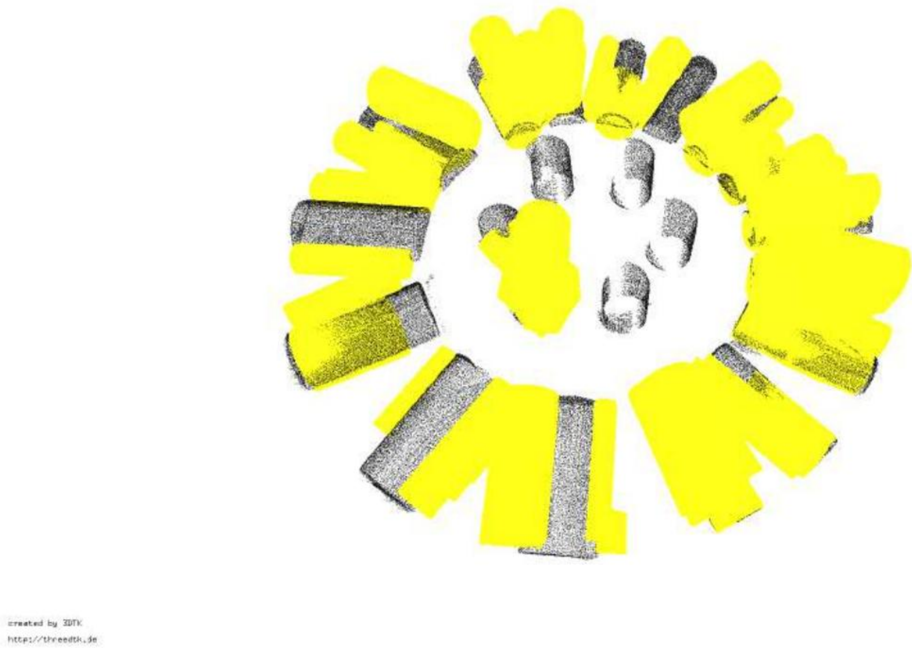


Figure 5.26: Erroneous detections in a real point cloud

However, the remaining 6.8% are not purely false positives. This is especially true for the half-tests. Horizontal cylinders with a radius of 5mm are often incompletely detected cylinders. whose cylinder height was determined to be too short or which have undergone a slight translation. Approximately 5% of the 21,000 cylinders, or 1,050 detections, are complete false detections, because they lie between two cylinders or matched to the remaining points of the standing cylinders All unsorted cylinders are shown in Figure 5.26. For a reduction To address false detections, the parameters of the circular validation function must be adjusted. For example, by using a finer circular step resolution or by increasing the number of segments, on which a point must lie. However, in the given case of real-world data, this is only very possible. That's difficult, since most cylinders are half-cylinders. A narrowing. This parameter would thus reduce the number of false detections, but at the same time the Increase the false detection rate.

Cylinder [Radius; Height]	Number	Mean Radius [mm]	Mean Axis
Cyl. 0-1 [5mm; [5, 10]mm]	-	-	-
Cyl. 2-5 [5mm; [15, 20, 25, 30]mm]	3997	4,977088	[0.001380 0.999189 0.006357]
Cyl. 6 [7.5mm; 30mm] 993 cyl. 7		7.427172	[-0.493197 -0.001885 0.869653]
[7.5mm; 30mm] 999 cyl. 8 [7.5mm;		7.526029	[-0.850309 0.007853 0.525317]
30mm] 996 cyl. 9 [7.5mm; 30mm]		7.426477	[0.999116 -0.008590 0.014987]
1000 cyl. 10[7.5mm; 30mm] 1000		7.686630	[0.878575 -0.004868 0.476879]
cyl. 11[5mm; 30mm]		7.757408	[0.512793 -0.002961 0.858181]
	927	5.234428	[0.002132 -0.004140 0.999379]
Cyl. 12 [5mm; 30mm]	989	5.108334	[-0.302530 0.000483 0.952829]
Cyl. 13 [5mm; 30mm]	994	5.076445	[-0.571366 0.004360 0.820178]
Cyl. 14 [5mm; 30mm]	831	4.905990	[-0.799127 0.010327 0.599457]
Cyl. 15 [5mm; 30mm]	877	4.867855	[-0.943658 0.008873 0.327434]
Cyl. 16 [5mm; 30mm]	978	4.897739	[0.998687 0.025507 0.015339]
Cyl. 17 [5mm; 30mm]	926	4.979150	[0.955972 -0.007939 0.289266]
Cyl. 18 [5mm; 30mm]	998	4.946013	[0.823721 -0.008050 0.566299]
Cyl. 19 [5mm; 30mm]	989	4.805956	[0.601837 -0.005098 0.798163]
Cyl. 20 [5mm; 30mm]	993	4.869633	[0.321836 0.000422 0.946461]
Cyl. 21 [5mm; 30mm]	889	4.909966	[0.017559 -0.002393 0.998978]

Table 5.22: Detected cylinders, as well as mean radius and mean cylinder axis

Table 5.22 lists the individual cylinders according to the number of times they were detected, with the stationary cylinders grouped into two sets. The first set consists of the two smallest cylinders, 0 (5 mm high) and 1 (10 mm high). Due to the large shadows cast by these cylinders, many points on their surface are missing. Therefore, like cylinder 0 in the simulation dataset, they can be detected using the circle validation method. Adjusting the validation parameters, for example by reducing the circle resolution step size, is not possible, as this would increase the number of false detections. The second set consists of cylinders 2, 3, 4, and 5 with heights of 15, 20, 25, and 30 mm, respectively. Cylinders 3-5 were detected 1000 times, while cylinder 2 was missed in only three runs. However, the accuracy of their axis, radius, and position is very similar, which is why they are often considered together in the following analysis.

Even the half-cylinders lying horizontally with a radius of 7.5 mm are detected in almost every run. Cylinders 9 and 10 were found 1000 times each, while cylinder 6, with seven false detections, was the most frequently missed. Thus, the half-cylinders with a radius of 7.5 mm exhibit a relatively low false detection rate. This rate increases significantly when considering the half-cylinders with a radius of 5.0 mm; for example, cylinder 13 was detected 831 times. It is noteworthy, however, that the cylinders on one axis, such as cylinders 11 and 21, have detection rates of 889 and 927, respectively. This is still below the average of 944,636 (or 956 without cylinder 13) finds, but is far less critical than in the simulation data. This problem described at the beginning is mitigated because the maxima in Hough space are significantly wider for cylinder axis detection, allowing more axes to be tested and increasing the probability of finding both cylinders.

Since cylinders 1 and 0 have too few lateral surface areas to be calculated with the selected validation parameters, they are not included in the following evaluation.
Therefore, without these two cylinders, a total of 20,000 cylinders should have been detected; the actual number found, 19,376, is 624 cylinders too low. Slightly more than half of these missing cylinders were considered false positives because at least one of their parameters showed excessive deviations. In general, Table 5.22 shows that lower numbers of surface points negatively impact the detection success. The following sections analyze the errors of the individual parameters and examine their distribution in more detail.

5.3.6 Evaluation of cylinder axis detection on real data

Since the data is real, a comparison of the identified axes with absolute values is not possible, just as with the division of the cylinders. Instead, the angles between the detected cylinder axes are calculated for evaluation purposes, as it is known that the angle between two half-cylinders with a radius of 5 mm is 18° and otherwise 30° .

Table 5.23 shows the individual differences of the mean angles and their errors from the actual value (30° or 18°) . The largest error is approximately 2.7° and the minimum -0.053°. Since the angle differences are already known for the axes in the plane, it is also useful to consider the angles of the axes to this plane. Projecting

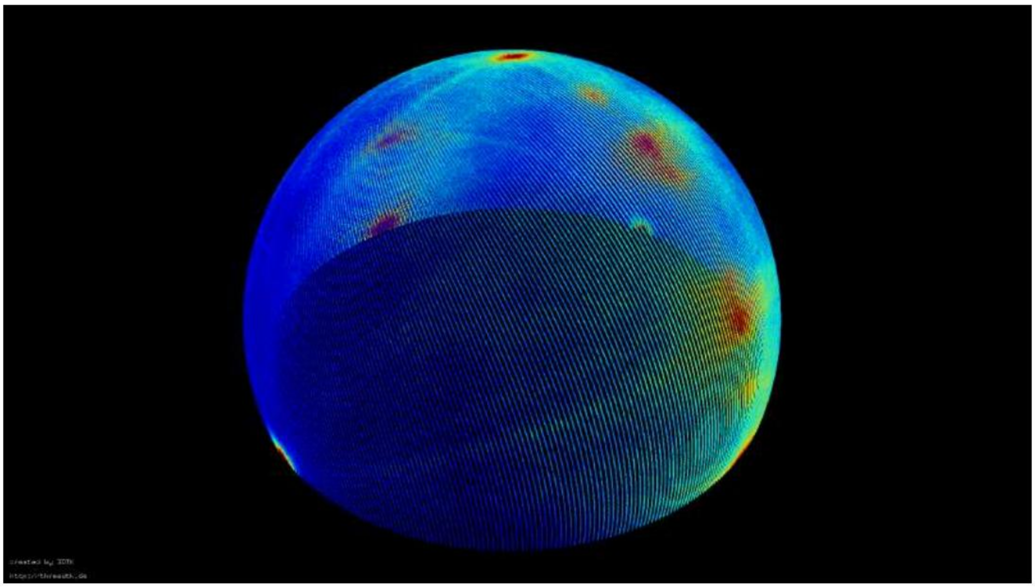


Figure 5.27: Representation of the accumulator for detecting the cylinder axes, where blue represents unincremented cells and red represents cells with maximum accumulation.

The axes along the central cylinder axis from 2-5, which represents an erroneous plane normal, and calculate the angle difference, the error range lies in [-2.25°, 2.68°] with the smallest deviation of -0.04°. The mean absolute error is 0.951°, which is only slightly smaller than the absolute error of the unprojected axes, which is 0.952°. The relative angles are not an absolute indicator of the errors of the individual axes, but they do indicate that they are at least within the correct range.

Since the absolute value of the axis is unknown, it is important to consider the distribution of the individual axes. For a very rough analysis, the representation of Hough space (see Figure 5.27) can be helpful. The points on the great circle represent the half-cylinders lying on their sides, while the vertical axis is shown at the edge of the sphere (bottom left in the image). It is clearly visible that the maxima are distributed to varying degrees, and some small maxima, which mostly originate from half-cylinders lying on their sides with a radius of 5 mm, lie at the edge of the large maxima. These large maxima arise, for example, from half-cylinders with a radius of 7.5 mm or from two opposing half-cylinders that share the same axis, which is why significantly more points on the lateral surface in Hough space vote for them.

If a small maximum lies at the edge of a larger maximum, it becomes significantly more difficult to detect and precisely determine it in the accumulator. This is because slightly erroneous spherical normals can also tune for a region around the local maximum, thus influencing the detection of a cylinder axis with only a few lateral points. Errors inherent in the real data exacerbate this problem, as spherical normals cannot be calculated exactly under these circumstances. For this reason, detecting small maxima at the edge of larger ones is considerably more difficult than, for example, detecting the clear maximum of the six standing cylinders.

Cylinder	Cylinder neighbor	Mean angle [ÿ]	Angle error [ÿ]
Cyl. 6	cyl. 7	28,805887	-1.194113
cyl. 7	cyl. 8	32.686509	2.686509
cyl. 8	cyl. 9	27.755328	-2.244672
cyl. 9	cyl. 10	30.707005	0.707005
cyl. 10	cyl. 11	30.823775	0.823775
cyl. 11	cyl. 12	17.925793	-0.074207
cyl. 12	cyl. 13	17.367793	-0.632207
cyl. 13	cyl. 14	18.921310	0.921310
cyl. 14	cyl. 15	17.850071	-0.149929
cyl. 15	cyl. 16	20.625440	2.625440
cyl. 16	cyl. 17	16.529179	-1.470821
cyl. 17	cyl. 18	17.947110	-0.052890
cyl. 18	cyl. 19	18.615572	0.615572
cyl. 19	cyl. 20	18.342506	0.342506
cyl. 20	cyl. 21	18.042419	0.042419
cyl. 21	cyl. 6	30.650542	0.650542

Table 5.23: Mean angle between projected cylinder axes

If one considers the dispersion of the individual axes, this phenomenon can be clearly seen. recognize. Figure 5.28 shows an example histogram for the axis of the Cylinder 9 is shown. Cylinder 9 is a half horizontal cylinder with a radius of 7.5 mm. It has This results in significantly more points than, for example, a horizontal cylinder with a radius of 5 mm. (See histogram.) It is clearly evident that the scattering is limited to a very small area and is more of a It has a linear structure. In contrast, the axis of cylinder 17 is plotted in 5.29. A peak is visible here, but it is much more widely distributed. The axes of the Cylinder 17 has a significantly larger dispersion range compared to cylinder 9. A similar The remaining detected cylinder axes also exhibit this distribution. These are not always Peak-shaped like cylinder 17, however the axes of the half horizontal cylinders are clearly further dispersed.

Considering the overall result of the cylinder axis detection, one can conclude that that the individual cylinder axes were essentially detected well. The angles The values between the individual cylinders remain unchanged. Although this is real data, it was Cylinder axis detection is possible for half cylinders, however the spread of the axes is significantly larger, especially for half-cylinders with small radii.

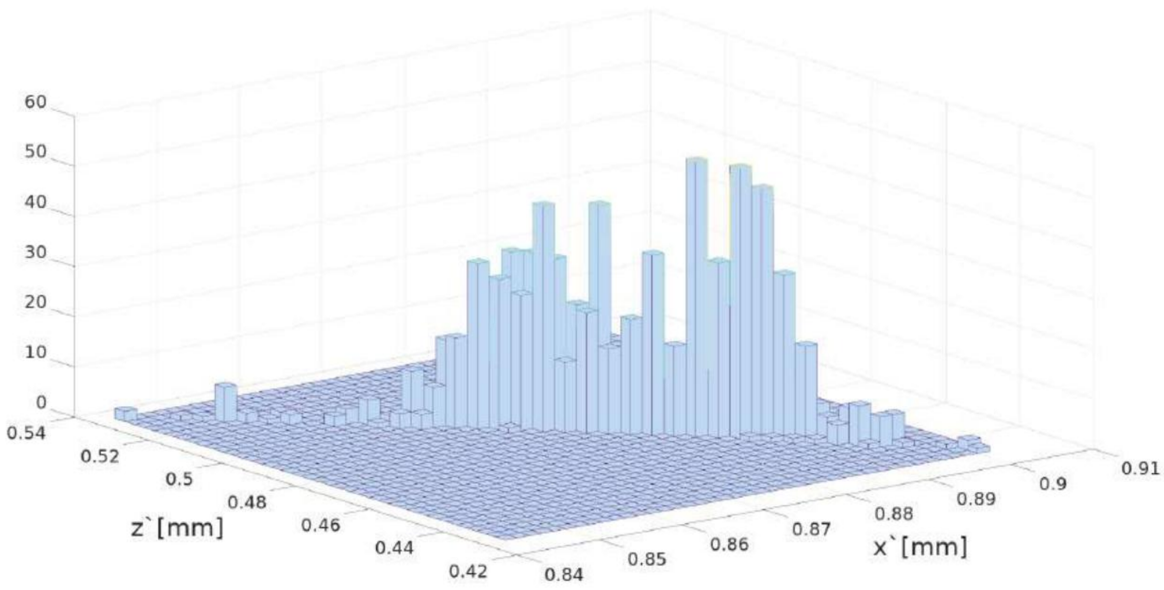


Figure 5.28: Histogram of the deviation of the nominal axis of cylinder 9

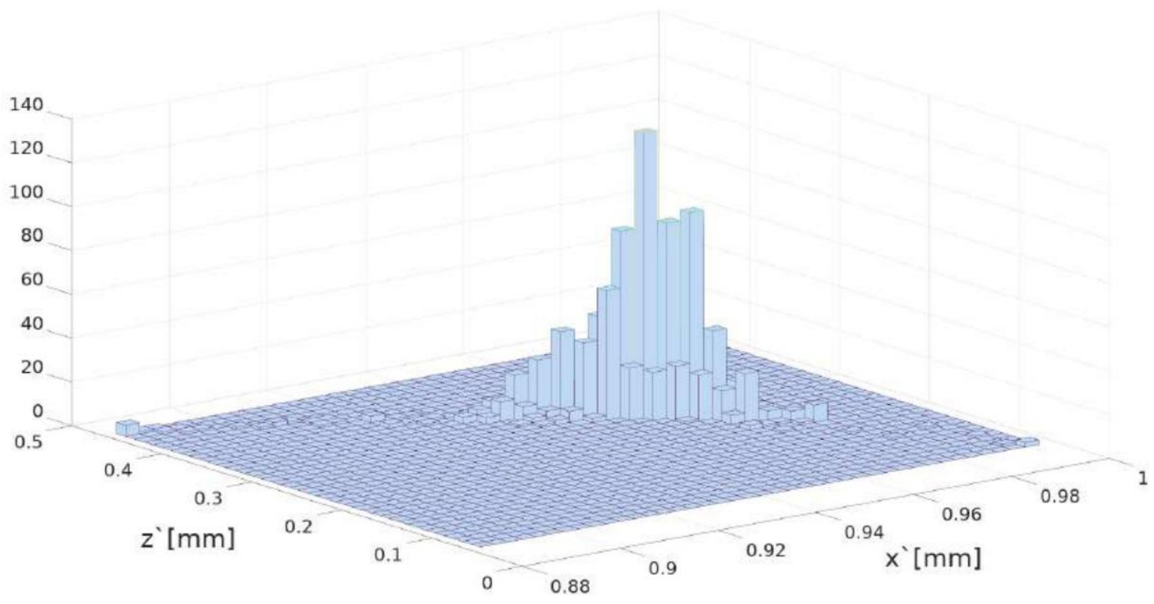


Figure 5.29: Histogram of the deviation of the target axis of cylinder 17

5.3.7 Evaluation of circle detection on real data

As already mentioned in section 5.3.5, the decisive limiting factor is not cylinder axis detection, but circle detection, provided no large planes are present.
Validation parameters that determine whether a set of points P, which lies on a circular path K,
Whether or not it represents a circle must be determined based on the half-cylinders in comparison to section 5.2.
This will be adjusted. In this case, this is done by lowering the positive threshold.
from 50% to 25% of all circle segments that must contain a point. For this reason
The number of false detections increases significantly, as P on K is more frequently recognized as a circle.
even if P is only a random set of points. At the same time, half of them possess
Cylinders, especially those with a small surface area, have an increased false detection rate compared to other cylinders.
to solid cylinders. Furthermore, the detected semicircles have a significantly increased uncertainty.
which is why the fluctuations of the circle's center point and radius increase significantly.

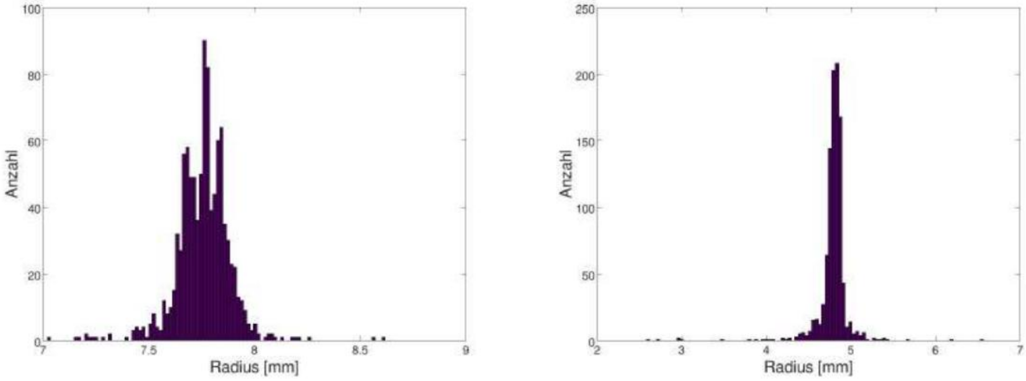
This phenomenon can be clearly observed using the radii shown in Table 5.24, as well as their standard deviation. While the mean radius of the whole cylinders is a
exhibiting a deviation of 0.022912mm and a standard deviation of only 0.071107mm, the mean radius of the half cylinders with the significantly larger radius of 7.5mm
an error of -0.059713 mm and the standard deviation is about 4.6 times larger than the
of the entire cylinders. Looking at the half-cylinders with a radius of 5.0 mm, it becomes apparent that...
an even greater difference. The error of the mean value here is already 0.094329 mm.
while the standard deviation of 0.432650 is already 6 times larger than that of the
standing whole cylinder with radius 5mm.

cylinder	Radius [mm]	Medium radius [mm]	Standard deviation [mm]
Entire cylinders	5	4.977088	0.071107
Half cylinders	7.5	7.559713	0.329613
Half cylinders	5	4.905671	0.432650

Table 5.24: Comparison of mean radii and standard deviations

If we consider the radii with the smallest standard deviation, these are, in the case of half the
Horizontal cylinder with radius 7.5mm, cylinder number 10, and for a radius of 5mm
Number 18. It is noticeable that both cylinder 10 and cylinder 18 have very high...
They exhibit detection rates of 1000 and 998, respectively. Figure 5.30 shows the individual...
The detected radii of these two cylinders are shown. Already when comparing the
The histogram clearly shows, based on the x-axis, that cylinder 18 has a significantly wider range.
The distribution shows 10 cylinders.

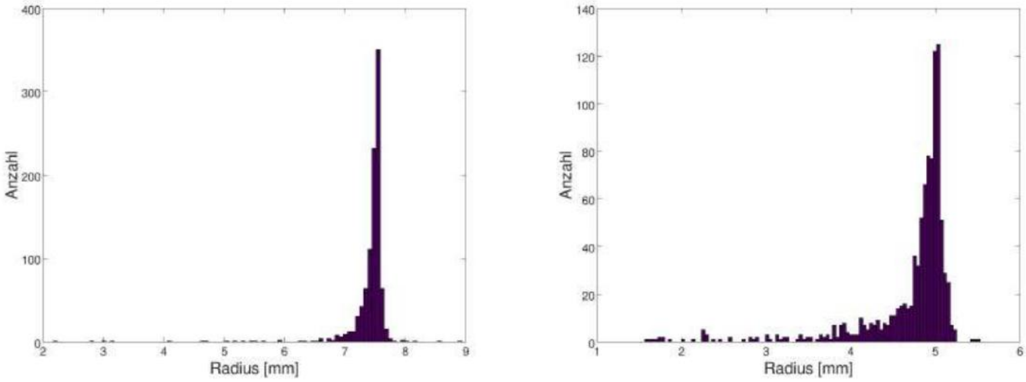
This observation is reflected even more strongly in the radius ranges of the individual cylinders.
again, while cylinder 10 has a small range of [7.0202 8.6189], the
The range of values for the radii of cylinder 18 is significantly more critical at [1.5610 5.5301]. In contrast, the
The median of the small half-cylinder 18, at 4.9184 mm, is better than the median of the cylinder 10.
7.7651 mm. The median clearly shows that there are only a few lateral surface points.



(a) Histogram of the 1000 detected radii of the Cylinder 10

(b) Histogram of the 1000 detected radii of the Cylinder 18

Figure 5.30: Histograms of the radii of cylinders 10 (left) and 18 (right)



(a) Histogram of the 1000 detected radii of the Cylinder 6

(b) Histogram of the 1000 detected radii of the Cylinder 15

Figure 5.31: Histograms of the radii of cylinders 6 (left) and 15 (right)

These parameters can be used for circle detection, resulting in either a very good or no fit. In the special case of half cylinders, a fit with a very small radius, tangent to the actual cylinder surface, can also occur. The latter was made possible by lowering the circle detection validation parameters. Due to the very small radius of 1.5610, only *0.25* × 395 = 99 segments need to contain a point, which can occur due to outliers and points on the original surface of the cylinder. Besides visual observation, the standard deviation describes the distribution of the detected radii of cylinders 10 and 18, and thus their difference, very well. While cylinder 10 has a standard deviation of 0.12821, cylinder 18, at 0.21373 mm, has a standard deviation almost twice as large.

However, the standard deviation of cylinder 18 is smaller than that of cylinder 6, although this Cylinder 6 also has a radius of 7.5 mm. Cylinder 6 exhibits the largest standard deviation of all half-cylinders with a radius of 7.5 mm, at 0.42250 mm. This can be clearly seen in Table 5.25. can be read off, showing the individual medians, mean radii and standard deviations. The errors are plotted. However, cylinder number 15 has the greatest variance with a standard deviation of 0.585470 mm. For this reason, Figure 5.31 shows... In addition to the histogram of the found radii of cylinder 6, the radii of cylinder 15 are also shown. This representation clearly shows that a very large number of values are applied to cylinder 15. Small and, in some cases, significantly too small radii were found. Nevertheless, even for The worst detected cylinders show a clear peak in the actual radius area can be found.

cylinder	Radius [mm]	Median [mm]	Mean Radius [mm]	Standard deviation [mm]
Cyl. 6	7.5	7.499560	7.408533	0.422502
cyl. 7	7.5	7.588930	7.526029	0.246862
cyl. 8	7.5	7.453545	7.419194	0.398608
cyl. 9	7.5	7.718365	7.686630	0.173870
cyl. 10	7.5	7.765100	7.757408	0.128208
cyl. 11	5.0	5.257740	5.146516	0.577249
cyl. 12	5.0	5.143900	5.102622	0.182425
cyl. 13	5.0	5.086810	5.067561	0.201829
cyl. 14	5.0	4.875085	4.771043	0.566478
cyl. 15	5.0	4.918440	4.711516	0.585470
cyl. 20	5.0	4.915200	4.873486	0.336596
cyl. 16	5.0	5.042520	4.906609	0.441586
cyl. 17	5.0	4.955030	4.943652	0.306659
cyl. 18	5.0	4.810680	4.791209	0.213731
cyl. 19	5.0	4.897810	4.864450	0.138246
cyl. 21	5.0	4.885380	4.771152	0.578987

Table 5.25: Mean radii and standard deviations of the half cylinders

Evaluation of circle detection - point on cylinder axis

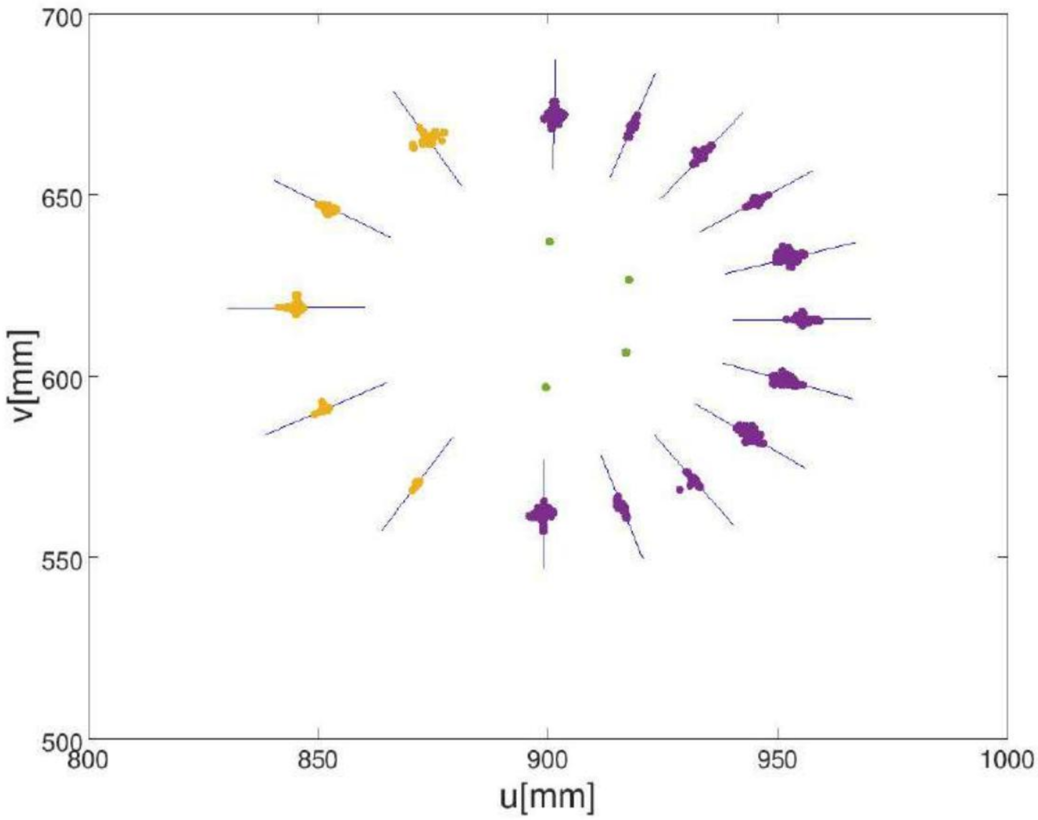


Figure 5.32: Representation of the cylinder axis centers projected onto the model plate

Since the second sequential step not only determines the radius but also identifies a point on the cylinder axis to determine the cylinder's 3D position in space, this point is evaluated in the following section. The disadvantage here is that, as with the cylinder axis, there are no absolute values for these points. Furthermore, not all points can be projected along the cylinder axis for evaluation, as this would require evaluating 15 planes, and two pairs of cylinders would lie on the same circle. Therefore, instead of the determined point on the cylinder axis itself, the midpoint between the cylinder's start and end points is calculated and compared.

This is possible because the start and end points are calculated from the detected circle center. However, the calculation also uses the cylinder axes and the lateral surface points used for circle detection. Therefore, the error of the cylinder axis center also affects the error of the cylinder axis itself and is not a pure indicator of the quality of the circle center.

Figure 5.32 shows the calculated cylinder axis centers projected onto the plane.
described by the central cylinder axis of cylinders 2-5. The blue lines represent the
The projected mean values of the cylinder axes are represented. The cylinder centers lie on these axes.
where green represents the points of the whole cylinders, purple those of the halves with a radius of 5.0 mm, and yellow the
The half-axis with a radius of 7.5 mm is marked. The error of the projected cylinder centers
This is represented by the distance to the cylinder axis. However, this should always be taken into account.
This means that the cylinder axes are also defective. The greater this defect,
The less the distance reveals about the actual errors of the axis points, but rather
only one indicator for the dispersion remains. However, since the points are around the center of the axis...
are arranged and the relative angles between the axes are maintained, is used for the
Evaluation assumes that the error is present, but small enough to at least
to be able to make a relative statement about the errors of the circle centers.

If one considers the cylinder axis centers in Figure 5.32, one can already see visually
It is very good that the points of the half-cylinders are significantly more widely scattered than the points of the
The whole. While the green dots appear visually to be a single point, it is recognized
There are a few individual yellow dots. Compared to the half-cylinders with a radius of 7.5 mm, they are
the cylinder axis centers of the half-cylinders with a radius of 5mm, however, again
significantly more widely dispersed. Nevertheless, it is always clearly visible that the points are concentrated on or
move in close proximity to the respective central cylinder axis.

cylinder	mean projected Axis point [mm]	minimal Distance [mm]	maximum Distance [mm]	Standard deviation [mm]
Cyl. 2	[899,551061 596.935052]	0.001879	0.183149	0.016280
Cyl. 3	[900,414233 637,145877]	0.001738	0.164206	0.016312
Cyl. 4	[917,017687 606,568438]	0.000923	0.119993	0.020193
Cyl. 5	[917,632698 626,609747]	0.001434	0.125434	0.017860

Table 5.26: Error distance of the cylinder axis centers of the entire cylinders

This phenomenon can also be observed when considering the maximum distance and the
Standard deviation between the whole cylinders in Table 5.26, with the mean distance
comparing the half cylinders in Table 5.27, it is clearly evident that the standard deviations of the half cylinders
are at least 10 times larger than those of the whole cylinders. Since the
The average distance between the half cylinders is significantly larger in most cases, and in very few cases
approximately equal to the maximum distance between the projected vertical cylinder axis centers,
It is also clear that half circles are detected significantly worse than whole circles.

Considering that half cylinders represent a borderline area of the partial cylinders to be detected
This is acceptable. The positions of the cylinders with a radius of 7.5 mm are particularly suitable.
a maximum standard deviation of 0.414 mm and a maximum mean distance of
0.335mm was reliably detected without significant variation. Therefore, it is particularly suitable for a
Semicircle detection is important, that a cylinder detects as many as possible, preferably evenly distributed.
possesses lateral surface points.

cylinder	Median distance [mm]	Mean distance [mm]	Standard deviation [mm]
Cyl. 6	0.139363	0.213163	0.414009
cyl. 7	0.134753	0.194845	0.229361
cyl. 8	0.233345	0.334971	0.367996
cyl. 9	0.093477	0.147497	0.175013
cyl. 10	0.101387	0.126727	0.107072
cyl. 11	0.321996	0.483561	0.501750
cyl. 12	0.110031	0.142013	0.172677
cyl. 13	0.103925	0.144636	0.252326
cyl. 14	0.177189	0.403198	0.532838
cyl. 15	0.358673	0.498603	0.643388
cyl. 20	0.178431	0.274297	0.313187
cyl. 16	0.220888	0.344923	0.491740
cyl. 17	0.108234	0.184020	0.301768
cyl. 18	0.090324	0.150751	0.221282
cyl. 19	0.072281	0.102173	0.130615
cyl. 21	0.343605	0.496063	0.505122

Table 5.27: Error distance of the projected cylinder axis centers of the half cylinders

5.3.8 Evaluation of partial cylinder detection on real data

Looking at the results of the real data, it can be stated that partial cylinders
can be found using the algorithm presented in this work, as long as it does not
There are many outliers and a small number of false detections is not critical.
However, half cylinders are determined with a significantly greater uncertainty and can
They are often not detected, especially with very small surface areas. The radius of a
The half cylinder is often detected as being too small.

A total of 20,796 cylinders were detected. Of these, 19,376 were correctly located. The remaining 7% are divided into false detections (5%) and similar cylinders that, however, have at least one incorrect parameter (2%). Complete false detections are very rare and are primarily associated with cutting-edge false detections.

In summary, it is possible to detect partial cylinders down to half-cylinders, but the number of false detections increases rapidly. By defining conditions that aid circle detection, it is possible to minimize the number of false detections. If partial cylinders are to be detected on real data with more outliers, the number of points sampled on the surface should be as precise as possible.

5.4 Summary of Cylinder Detection Results

In summary, the experiments show that the cylinder detection approach works.

Considering the evaluation of the noise-free point clouds, the experiments for evaluating circle detection with an erroneous axis (5.1.2), the simulated cylinder detection experiment (5.2), and the simulated half-cylinder detection experiment (5.3) required the detection of a total of 16,000 + 21,000 + 6,000 = 43,000 complete cylinders. The algorithm found a total of 15,999 + 20,895 + 4,993 = 41,887 cylinders, which corresponds to approximately 97.4% of all cylinders. It is important to consider that a standing cylinder in Model 3 has a surface area that is too small and was therefore not detected in any of the 1,000 trials due to the validation function of the circle detection.

Thus, only 113 false detections due to unexplained random errors were recorded. If the systematic errors of the 1000 false detections are excluded, then a total of 99.7% of all cylinders were detected.

For the detection of half-cylinders on noise-free point clouds, 16,000 half-cylinders were to be detected in Section 5.3. However, only 15,698, or approximately 98.1% of all cylinders, were found. Almost all false detections can be attributed to a boundary condition: If two cylinders have the same radius and the same axis, and their position in space is identical except for a slight shift along the axis, then the cylinders projected along their axis form the same circle on the plane. As shown in Section 5.3, this can lead to false detections if the surface points of the first cylinder are not largely detected and deleted. Only a single false detection originated from a cylinder for which this boundary condition did not apply.

The false detections for the cylinders in both the simulated and real data for Model 2 typically have a very small radius. Furthermore, the falsely detected cylinders exhibit a cylinder axis that usually deviates by at least 45° from the true cylinder axes. This false cylinder axis is calculated in the first step of cylinder detection based on the plane points. These points also lead to the detection of cylinders with very small radii that are tangent to the plane points during circle detection.

To avoid such false detections in the future, planes must be completely removed from the point cloud beforehand, or the evaluation of the Hough space for cylinder axis detection must be improved, and the validation function for circle detection must be adapted by further restrictions.

However, the false detections in the real data for Model 3 reveal a further weakness. While it is possible to find partial cylinders with the algorithm presented in this work, the conditions for the validation function of the circle detection must be relaxed, resulting in an increased number of false detections. Furthermore, the detection rate of half cylinders is significantly lower than that of whole cylinders in real data.

In the 1000 detections of model 3, 15,379 out of 16,000 half cylinders were detected (approximately 96.1%), while 3,997 out of 4,000 (6,000) full cylinders were detected. Ideally, 6,000 full cylinders should also have been detected; however, the cylindrical surface area of two cylinders is too small due to shadowing. If the detected cylinders of model 2 are also taken into account, a total of 20,858 + 3,997 = 24,855 out of 25,000 (27,000), or approximately 99.4% (92.0%) of all cylinders, were successfully detected.

Overall, the evaluation clearly shows that the cylinder detection method developed in this work offers some significant improvements in runtime and accuracy compared to that described in [22], with no restrictions on the cylinder parameters. Therefore, it is not necessary to define a radius range for building the accumulator. At the same time, the evaluation also reveals some boundary conditions under which the algorithm fails, for example, due to an insufficient lateral surface area or a degraded result. The following outlook will discuss possible extensions that could minimize the problems mentioned in the evaluation and further improve cylinder detection.

Chapter 6

Summary and Outlook

In summary, this work presents a randomized Hough transformation that makes it possible to detect cylinders and sub-cylinders.
For this purpose, a sequential two-step approach is used, whereby first the cylinder axis and then the radius and the position in three-dimensional space are determined.

Due to the randomization, the algorithm is significantly less computationally intensive than a standard Hough transformation approach, as presented in the paper on which it is based [22].
At the same time, the combined RHT and Least-Square approach of the second sequential step of circle detection also improves the runtime, while maintaining the quality of the circle detection.

Furthermore, the approach presented in this work does not impose any requirements on the cylinder parameters. Thus, the axis, radius, and position of the cylinders can all be freely chosen. However, it is possible to incorporate further conditions into the presented algorithm, thereby improving its runtime and robustness. Especially when detecting partial cylinders, particularly half-cylinders, false detections can occur more frequently if no conditions are present. For this reason, it is advisable to consider the cylinder detection algorithm presented in this work as an easily extensible foundation that can be optimized for specific problems as needed.

The following is a brief outlook on possible useful extensions and solutions to the problems encountered in this work, so that cylinder detection in large point clouds will also be possible in the future.

The biggest problem with this approach lies in its high susceptibility to errors when planes are present in the point cloud. As a solution, this paper, like [22], proposes performing plane detection before the actual cylinder detection. This can be useful, as in [15]. In the special case presented there, conditions for the cylinder axis could be established by detecting the ceiling and the floor, since all cylinders are perpendicular or parallel to the planes. However, the situation is different if no further conditions can be derived from the plane detection. In this case

While plane detection is necessary for the algorithm's functionality, it is not particularly useful. In such a case, it is actually more sensible to filter out only very large planes beforehand and to make the evaluation of the Hough space for cylinder detection more stable by considering only point-like peaks and not circular or line-shaped maxima in the accumulator as the cylinder axis.

Secondly, the implementation should be extended to minimize false detections that occur when two cylinders have the same radius and axis but differ in their spatial positions due to translation along the axis. The simplest way to avoid this is to introduce an additional threshold that ensures that, after the successful initial detection of one of the two cylinders, not only the directly used points are deleted, but also all points within a certain distance of the cylinder. However, this threshold would vary considerably from point cloud to point cloud. Another way to circumvent this problem is through a more precise implementation of the cylinder end search, for example, using a DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm. This algorithm first calculates all points of the cylinder on its surface and then deletes them completely, or at least sufficiently, so that no contiguous cylinder can be found in the second step. Furthermore, a DBSCAN approach could be applied to the entire point cloud after cylinder validation to correct the systematic error caused by excessively short cylinder lengths.

The aforementioned extensions are intended to serve as a basis for highlighting the errors encountered in this work and offering a solution for possible future implementations. These extensions should enable robust, fast, and reliable cylinder detection even in larger point clouds with more outliers.

The goal of a randomized Hough transform for cylinder detection in laser scans was successfully implemented in this work. Even the detection of partial cylinders down to half-cylinders is possible, as long as a small number of false detections is not critical or further conditions are met to minimize them.

Attachment

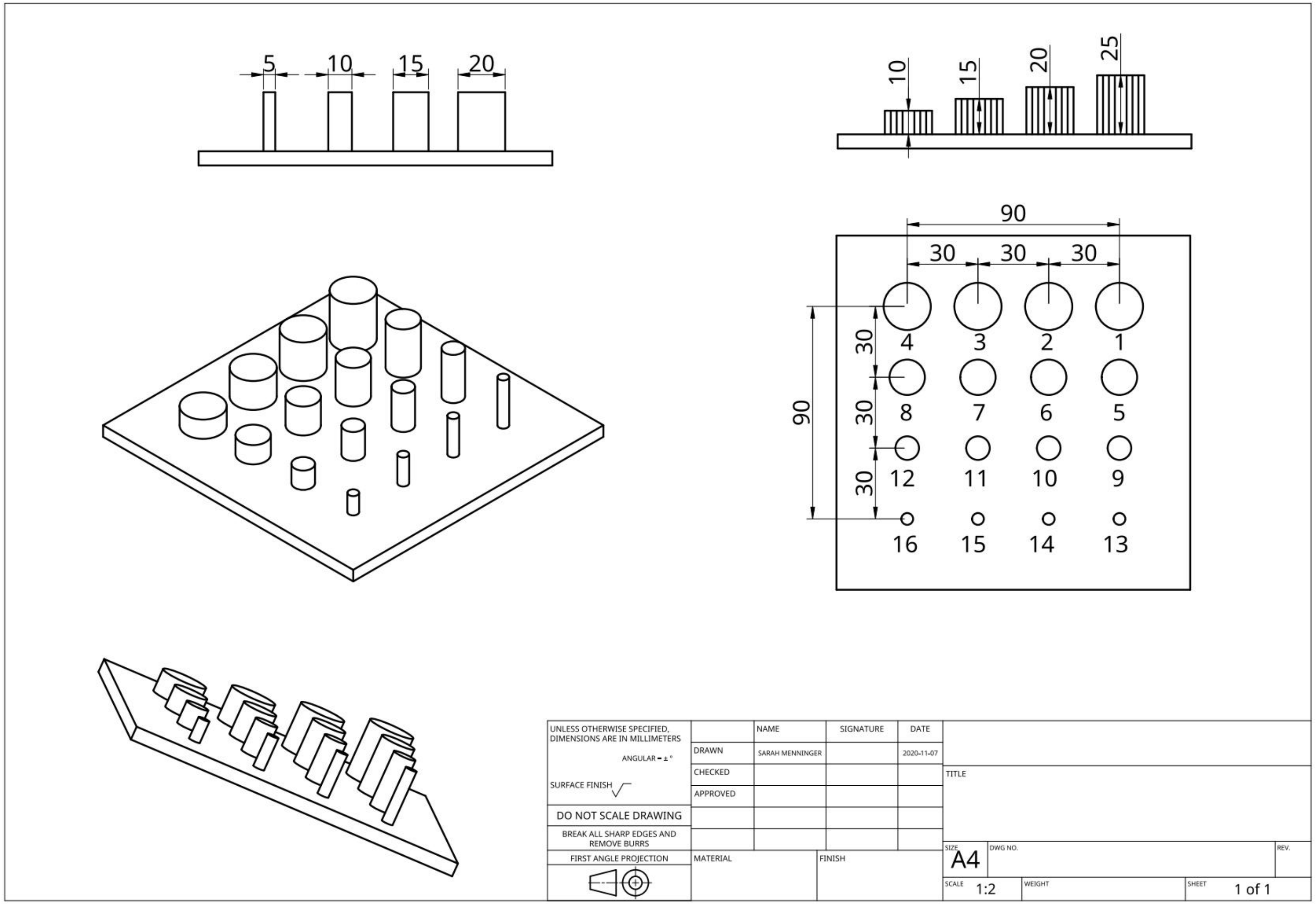


Figure 6.1. Representation of the 1st model for evaluating circle detection

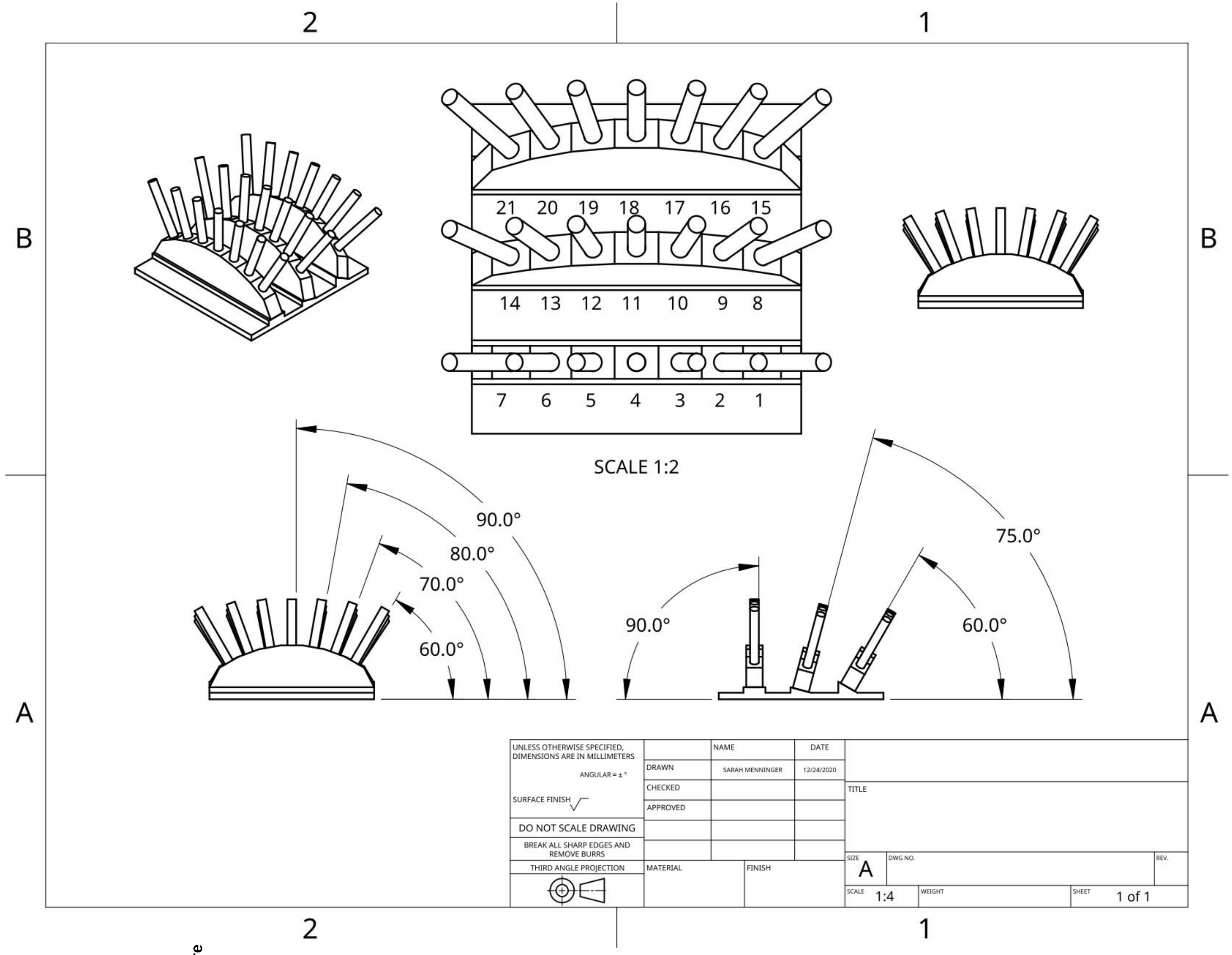


Figure 6.2: Representation of the 2nd model for evaluating the detection of cylinders

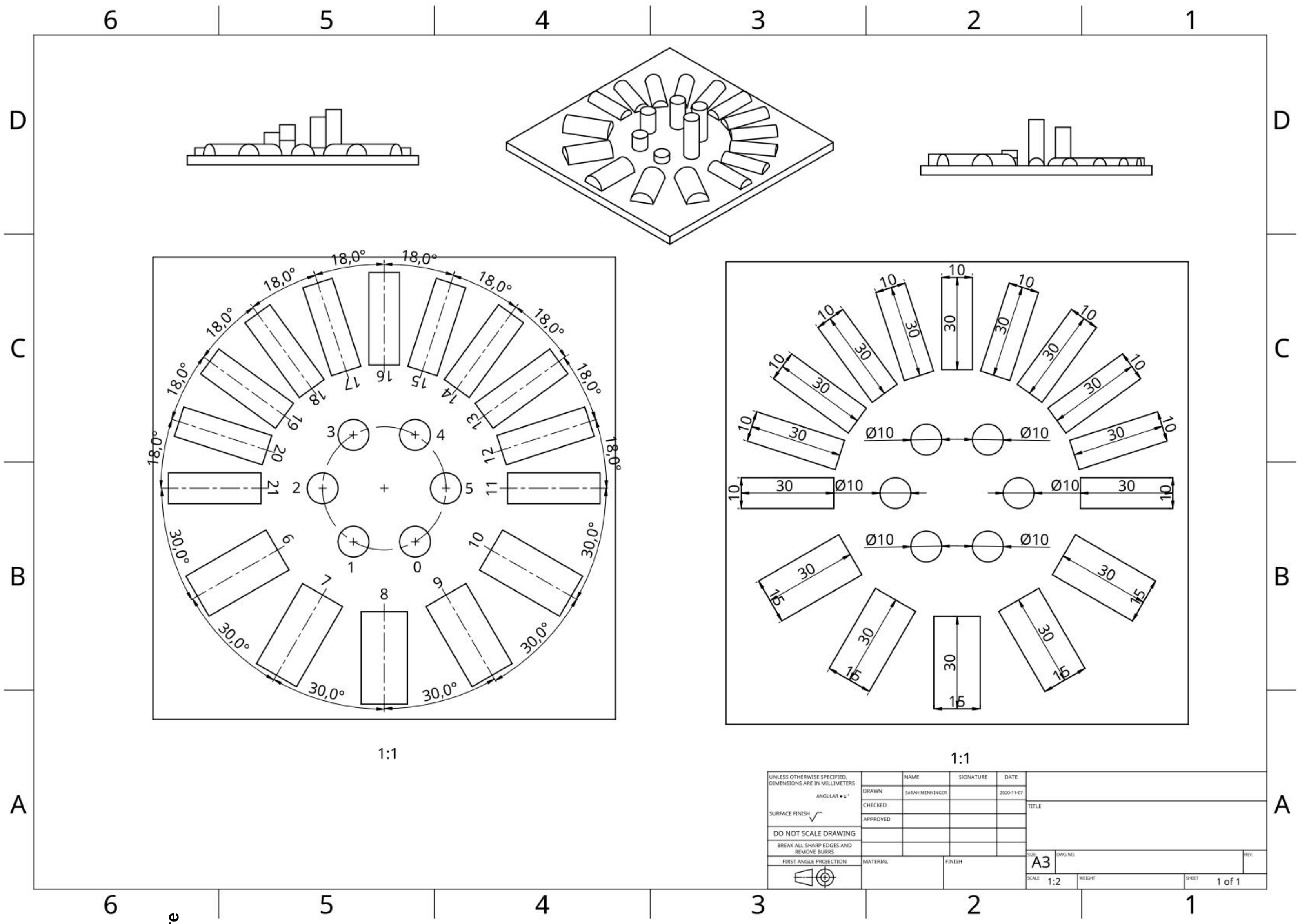


Figure 6.9: Representation of the 3rd model for evaluating the detection of half cylinders

Bibliography

[1] *3DTK - The 3D Toolkit.* http://slam6d.sourceforge.net/, . – Accessed: 2020-11-23

[2] *CloudCompare.* https://www.danielgm.net/cc/, . – Accessed: 2020-12-6

[3] *Onshape.* https://www.onshape.com/en/, . – Accessed: 2020-12-6

[4] *scanCONTROL 2900-25.* https://www.micro-epsilon.de/2D_3D/laser-scanner/scanCONTROL-2900/, . – Accessed: 2020-12-8

[5] *University of Würzburg - Computing Centre - 3D scanning.* https://www.rz.uni-wuerzburg.de/services/multimedia/3d-services/3d-scanning/, . – Accessed: 2020-12-2

[6] Badino, H.; Huber, D.F.; Park, Y.; Kanade, T.: Fast and accurate computation of surface normals from range images. In: *Proceedings - IEEE International Conference on Robotics and Automation* (2011), 06, pp. 3084 – 3091

[7] Borrmann, D.; Elseberg, J.; Lingemann, K.; Nüchter, A.: The 3D Hough Transform for plane detection in point clouds: A review and a new accumulator design. In: *3D Research* 2 (2011), No. 2

[8] Chaperone, T.; Goulette, F.: Extracting Cylinders in Full 3D Data Using a Random Sampling Method and the Gaussian Image. In: *Proceedings of the Vision Modeling and Visualization Conference* (2001), 01, pp. 35–42

[9] Golub, G. H.; Loan, CFV: *Matrix Computations.* Vol. 4. Baltimore, Maryland 21218-4363: The Johns Hopkins University Press, 2013

[10] Guo, S.; Zhang, X.; Zhang, F.: Adaptive Randomized Hough Transform for Circle Detection using Moving Window. In: *2006 International Conference on Machine Learning and Cybernetics,* 2006

[11] In: Henning, B.; Besslich, PW: *Hough transformation.* Berlin, Heidelberg: Springer, 1991, pp. 101–121

[12] Hoppe, H.; DeRose, T.; Duchamp, T.; McDonald, J.; Stuetzle, W.: Surface Reconstruction from Unorganized Points. In: *Proceedings of the 19th Annual Conference on Computer Graphics and Interactive Techniques.* New York, USA : Association for Compu-ting Machinery, 1992

[13] Kimme, C.; Ballard, D.; Sklansky, J.: Finding Circles by an Array of Accumulators. In: *Communications of the ACM* 18 (1975), No. 2, pp. 120 – 122

[14] Kiryati, N.; Eldar, Y.; Bruckstein, AM: A probabilistic Hough transform. In: *Pattern Recognition* 24 (1991), No. 4, pp. 303 – 316

[15] Liu, Y.; Zhang, J.; Hou, J.; Ren, J.; Tang, W.: Cylinder Detection in Large-Scale Point Cloud of Pipeline Plant. In: *IEEE Transactions on Visualization and Computer Graphics* 19 (2013), No. 10, pp. 1700–1707

[16] Lukács, G.; Martin, R.; Marshall, D.: Faithful Least-Squares Fitting of Spheres, Cylinders, Cones and Tori for Reliable Segmentation. In: *Proceedings of the 5th European conference on computer vision,* 1998

[17] Lutton, E.; Maitre, H.; Lopez-Krahe, J.: Contribution to the determination of va-nishing points using Hough transform. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 16 (1994), No. 4, pp. 430–438

[18] In: Marschner, S.; Shirley, P.: *Fundamentals of Computer Graphics.* 3rd. USA: AK Peters, Ltd., 2009, pp. 25–30

[19] Nurunnabi, A.; Sadahiro, Y.; Lindenbergh, R.: ROBUST CYLINDER FITTING IN THREE-DIMENSIONAL POINT CLOUD DATA. In: *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* XLII-1/W1 (2017), 05, pp. 63–70

[20] Papula, L.: *Mathematical Formula Collection - For Engineers and Natural Scientists.* Vol. 9. Wiesbaden: Viewegs Fachbücher der Technik, 2006

[21] Press, WH; Teukolsky, S.A.; Vetterling, W.T.; Flannery, BP: *Numerical recipes - The Art of Scientific Computing.* Vol. 3. The Edinburgh Building, Cambridge CB2 8RU : Cambridge University Press, 2007

[22] Rabbani, T.; Heuvel, F.: Efficient Hough transform for automatic detection of cylinders in point clouds. In: *Proc ISPRS Workshop Laser Scan 2005, ISPRS Arch* 36 (2005), 01

[23] Schnabel, R.; Choice, R.; Klein, R.: Efficient RANSAC for Point-Cloud Shape Detection. In: *Computer Graphics Forum* 26 (2007), 06, No. 2, pp. 214–226

[24] Taubin, G.: Estimation of planar curves, surfaces, and nonplanar space curves defined by implicit equations with applications to edge and range image segmentation. In: *IEEE Transaction on Pattern Analysis and Machine Intelligence* 13 (1991), 11, No. 11, pp. 1115–1138

[25] Thomas, SM; Chan, YT: A simple approach for the estimation of circular arc center and its radius. In: *Computer Vision, Graphics, and Image Processing* 45 (1989), No. 3, pp. 362 – 370

Erklärung

Hiermit versichere ich, dass ich die vorliegende Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie Zitate kenntlich gemacht habe.

Würzburg, Januar 2021